SGA 4 Translation Project – translation segment

Exposé XVII completion and Exposé XVIII opening through Section 2

Draft English LaTeX translation; compile-checked and rendered.

SGA 4, Exposé XVII – continuation

Cohomology with Proper Supports and the Construction of the Functor $f_!$

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this translation segment continues at the proof following Lemma 5.2.4. The statement of Lemma 5.2.4 was already included in the preceding translation segment; it asserted the compatibility of the base-change isomorphism for a composite $f = uv$ with the base-change isomorphisms attached separately to u and to v.

5.2. The base-change theorem, continued

Proof of Lemma 5.2.4. Choose a compactification of the pair (u, v). Thus one has a diagram

$$\begin{tikzcd}
X' \arrow[r, "g''"'] \arrow[d, "v'"'] & X \arrow[r, hook, "j_1"] \arrow[d, "v"] & \bar{X} \arrow[r, hook, "j_2"] \arrow[d] \arrow[dl, "p_1"'] & \bar{\bar{X}} \arrow[dl, "p_3"] \\
Y' \arrow[r, "g'"'] \arrow[d, "u'"'] & Y \arrow[r, hook, "j_3"] \arrow[d, "u"] & \bar{Y} \arrow[dl, "p_2"] & \\
S' \arrow[r, "g"'] & S & &
\end{tikzcd} .$$

The assertion is that the base-change isomorphism c_1 for f is the composite c_2 of the two isomorphisms for u and v. The isomorphism c_1 is obtained by composing the base-change isomorphisms attached to j_1, j_2, p_3, and p_2. The second isomorphism c_2 is obtained by composing those attached to j_1, p_1, j_3, and p_2. Hence the comparison reduces to the middle square: compare the composite attached to p_3 and j_2 with the composite attached to j_3 and p_1.

After the base change by g we obtain the Cartesian diagram

$$
\begin{tikzcd}
\bar{X}' \arrow[rrr, hook, "\bar{g}''"] \arrow[dr, hook, "j_2'"] \arrow[dd, "p_1'"'] & & & \bar{X} \arrow[dr, hook, "j_2"] \arrow[dd, "p_1"] & \\
& \bar{\bar{X}}' \arrow[rrr, "\bar{\bar{g}}"] \arrow[dd, "p_3'"] & & & \bar{\bar{X}} \arrow[dd, "p_3"] \\
Y' \arrow[rrr, hook, "g'"] \arrow[dr, hook, "j_3'"] \arrow[ddr, "u'"'] & & & Y \arrow[dr, hook, "j_3"] \arrow[ddr, "u"] & \\
& \bar{Y}' \arrow[rrr, "\bar{g}'"'] \arrow[d, "p_2'"] & & & \bar{Y} \arrow[d, "p_2"] \\
& S' \arrow[rrr, "g"'] & & & S.
\end{tikzcd}
$$

Put $L' = {p_2'}^* L$ and $K' = j_{1!} K$. The two composites to be compared are isomorphisms

$$
c_1', c_2' : L' \overset{\mathbf{L}}{\otimes} (\bar{g}')^* R(j_3 p_1)_! K' \xrightarrow{\sim} R(j_3' p_1')_! \Big((j_3' p_1')^* L' \otimes (\bar{g}'')^* K' \Big).
$$

By the elementary formula for extension by zero, it is enough to verify equality after restriction to the open $Y' \subset \bar{Y}'$, that is, after applying ${j_3'}^*$. On this open the assertion is immediate from the definitions of the base-change maps. This proves the compatibility for the chosen compactification. The usual comparison of two compactifications by a third one gives the result in general. $\square$

Lemma 5.2.5. Let

$$
(S'', \mathcal{A}'') \xrightarrow{g_2} (S', \mathcal{A}') \xrightarrow{g_1} (S, \mathcal{A})
$$

be morphisms of ringed schemes, the rings being sheaves of torsion rings, and let $f : X \to S$ be compactifiable. With the evident base-changed diagram

$$
\begin{tikzcd}
X'' \arrow[r, "g_2'"'] \arrow[d, "f''"] & X' \arrow[r, "g_1'"'] \arrow[d, "f'"] & X \arrow[d, "f"] \\
S'' \arrow[r, "g_2"'] & S' \arrow[r, "g_1"'] & S,
\end{tikzcd}
$$

where X, X', X'' are ringed by $f^*\mathcal{A}, f'^*\mathcal{A}', f''^*\mathcal{A}''$, respectively, the two ways of forming the base-change isomorphism for the composite $g_1 g_2$ agree. Equivalently, the diagram

$$\begin{tikzcd} L(g_1g_2)^*Rf_! \arrow[rr, "\sim"] \arrow[d, equal] & & Rf''_!L(g'_1g'_2)^* \arrow[d, equal] \\ Lg_2^*Lg_1^*Rf_! \arrow[r, "\sim"] & Lg_2^*Rf'_!Lg_1'^* \arrow[r, "\sim"] & Rf''_!Lg_2'^*Lg_1'^* \end{tikzcd}$$

is commutative.

Proof. The assertion follows immediately from the analogous statements for $j_!$ and for Rp_* when p is proper. One writes a compactification $f = pj$, expands the two composites, and uses functoriality of proper base change and of extension by zero. This is a purely formal pasting argument. □

Theorem 5.2.6. The functor $Rf_!$ commutes with base change.

This summarizes the preceding lemmas: the base-change morphism is defined independently of the chosen compactification, is compatible with composition of compactifiable morphisms, and is functorial for iterated changes of base.

Variants 5.2.7. The same assertions hold in the following forms.

(i) The functors $f_!$ for sheaves of pointed sets, and the functors $R^1f_!$ for sheaves of groups, commute with base change.

(ii) The variants of $Rf_!$ defined for bounded-below torsion complexes also commute with base change.

(iii) Under the hypotheses of 5.2.2, if L is bounded and has finite torsion dimension, then the functor $Rf_! : D(X) \to D(S)$ commutes with base change. As observed at the point where the morphism was constructed, the extra boundedness and finite-torsion-dimension hypotheses on L are not essential for the construction itself.

Theorem 5.2.6 is essentially equivalent to the conjunction of the following fibre formula and tensor formula.

Proposition 5.2.8. Let $f : X \to S$ be compactifiable, and let $\mathcal{F}$ be a torsion sheaf on X. For every geometric point s of S, if X_s is the geometric fibre and $\mathcal{F}_s$ the induced sheaf on it, there is a canonical identification

$$(R^q f_!\mathcal{F})_s = H^q_c(X_s, \mathcal{F}_s).$$

Corollary 5.2.8.1. If the fibres of f have dimension at most d, then $Rf_!$ has cohomological dimension at most $2d$: for every torsion sheaf $\mathcal{F}$, one has $R^i f_!\mathcal{F} = 0$ for $i > 2d$. Consequently $R^{2d}f_!$ is right exact on torsion sheaves.

Proof. By base change one reduces to the case where S is the spectrum of an algebraically closed field. Choose a proper compactification $j : X \hookrightarrow \bar{X}$ over S. Then $\dim \bar{X} = \dim X \le d$, and for a torsion sheaf $\mathcal{F}$

$$H^i_c(X, \mathcal{F}) = H^i(\bar{X}, j_! \mathcal{F}) = 0 \qquad (i > 2d),$$

by the cohomological dimension theorem for noetherian schemes. □

Proposition 5.2.9. Let $f : X \to S$ be compactifiable, let $\mathcal{A}$ be a sheaf of torsion rings on S, let K be a bounded-above complex of left $f^*\mathcal{A}$-modules, and let L be a bounded-above complex of right $\mathcal{A}$-modules. There is a canonical isomorphism

$$L \overset{\mathbf{L}}{\otimes}_{\mathcal{A}} Rf_!(K) \simeq Rf_!\Big(f^*L \overset{\mathbf{L}}{\otimes}_{f^*\mathcal{A}} K\Big).$$

Theorem 5.2.10. Let $f : X \to S$ be compactifiable, $\mathcal{A}$ a sheaf of torsion rings on S, and $K \in D^-(X, f^*\mathcal{A})$. If K has finite Tor-dimension $\le d$, then $Rf_!K$ has finite Tor-dimension $\le d$.

Proof. It is enough to verify that for every right $\mathcal{A}$-module L on S,

$$\mathcal{H}^{-i}(L \overset{\mathbf{L}}{\otimes} Rf_!K) = 0 \qquad (i > d).$$

By Proposition 5.2.9 the left-hand side is

$$R^{-i}f_!\Big(f^*L \overset{\mathbf{L}}{\otimes} K\Big).$$

The hypothesis on K gives $\mathcal{H}^{-i}(f^*L \overset{\mathbf{L}}{\otimes} K) = 0$ for $i > d$, and the assertion follows since applying $Rf_!$ to a complex with no cohomology below degree $-d$ cannot create such cohomology. □

In the special case of constant noetherian rings one has a slightly more general statement.

Theorem 5.2.11. Let $f : X \to S$ be a morphism of sites, let A be a ring, let C be its centre, and let $d \in \mathbb{Z}$. Assume:

(i) $Rf_* : D^+(X, C) \to D^+(S, C)$ has finite cohomological dimension;

(ii) A is right noetherian;

(iii) S has enough points.

For every $K \in D^-(X, A)$, if K has Tor-dimension $\le d$, then $Rf_*K \in D^-(S, A)$ has Tor-dimension $\le d$.

Proof. By the finite cohomological dimension hypothesis, f_* is derivable on unbounded derived categories in the required range. By the pointwise Tor-dimension criterion, it suffices to test the stalks of Rf_*K. Thus, for every point s of S, and every finitely generated right A-module L, it suffices to prove

$$H^{-i}(L \overset{\mathbf{L}}{\otimes} Rf_*K) = 0 \qquad (i > d).$$

As in the proof of Theorem 5.2.10 this follows if the natural map

$$L \overset{\mathbf{L}}{\otimes} Rf_*K \longrightarrow Rf_*(L \overset{\mathbf{L}}{\otimes} K)$$

is an isomorphism. Take a resolution L^* of L by free finitely generated right A-modules. The functors involved are way-out to the right; hence it suffices to prove the assertion for the individual components of the resolution. In that case L is finite free, and the assertion is immediate. □

Remarks 5.2.12.

(i) The hypothesis that S have enough points is probably unnecessary.

(ii) If the noetherian hypothesis on A is omitted, the theorem remains valid provided the functors $R^i f_*$ commute with filtered inductive limits.

(iii) The assumption that the coefficient sheaf of rings on S be constant is essential. A counterexample is obtained by taking S to be the spectrum of a strictly local discrete valuation ring, taking f to be the inclusion of the generic point η, taking $\mathcal{A}$ to have value $\mathbb{Z}/n\mathbb{Z}$ at η and $\mathbb{Z}/n^2\mathbb{Z}$ at the closed point s, with $n > 1$ prime to the residue characteristics, and taking the module sheaf $f^*\mathcal{A}$.

5.3. The comparison theorem and the finiteness theorem

Let X be a scheme of finite type over $\mathbb{C}$. Associated with X are the ét-site $X_{\text{ét}}$, the classical or transcendental site X_{cl}, and a morphism of sites

$$\epsilon : X_{\text{cl}} \longrightarrow X_{\text{ét}}.$$

If $j : U \to X$ is an open immersion, the square

$$\begin{array}{ccc} U_{\text{cl}} & \longrightarrow & U_{\text{ét}} \\ \downarrow & & \downarrow \\ X_{\text{cl}} & \longrightarrow & X_{\text{ét}} \end{array}$$

identifies the topos of U_{cl} with the open subtopos of X_{cl} inverse image of the open subtopos $U_{\text{ét}} \subset X_{\text{ét}}$. If $p : Y \to X$ is proper, then the associated continuous map $p^{\mathrm{an}} : Y^{\mathrm{an}} \to X^{\mathrm{an}}$ is proper and separated; by Chow's lemma this is reduced to the projective case.

Let $f : X \to S$ be a compactifiable morphism of schemes of finite type over $\mathbb{C}$, and write $f = pj$ with $j : X \hookrightarrow \bar{X}$ open and $p : \bar{X} \to S$ proper. The diagram of classical and ét-sites is

$$\begin{tikzcd}
X_{\mathrm{cl}} \arrow[rrr, "\epsilon"'] \arrow[dr, "j_{\mathrm{cl}}"] \arrow[dd, "f_{\mathrm{cl}}"'] & & & X_{\text{ét}} \arrow[dr, "j"] \arrow[dd, "f"' near start] & \\
& \bar{X}_{\mathrm{cl}} \arrow[rrr, "\epsilon"'] \arrow[dl, "p_{\mathrm{cl}}"] & & & \bar{X}_{\text{ét}} \arrow[dl, "p"] \\
S_{\mathrm{cl}} \arrow[rrr, "\epsilon"'] & & & S_{\text{ét}}. &
\end{tikzcd}$$

The functor $p_{\mathrm{cl}*} j_{\mathrm{cl}!}$ is the usual direct image with proper support $f_{\mathrm{cl}!}$. Since the question is local on S, extension by zero carries flasque sheaves on X_{cl} to sheaves on $\bar{X}_{\mathrm{cl}}$ acyclic for $p_{\mathrm{cl}*}$, and hence

$$Rf_{\mathrm{cl}!} = Rp_{\mathrm{cl}*} j_{\mathrm{cl}!}.$$

By the construction of $j_!$ one has a base-change isomorphism

$$\epsilon^* j_! \xrightarrow{\sim} j_{\mathrm{cl}!} \epsilon^*.$$

By proper base change, one has a morphism

$$\epsilon^* Rp_* \longrightarrow Rp_{\mathrm{cl}*} \epsilon^*.$$

Composing these gives a morphism, for torsion complexes,

$$\epsilon^* Rf_! \longrightarrow Rf_{\mathrm{cl}!} \epsilon^*.$$

As in Lemma 5.2.3, this morphism is independent of the chosen compactification of f, and it satisfies the expected transitivity formula.

Theorem 5.3.3 (comparison theorem). The morphism

$$\epsilon^* Rf_! \longrightarrow Rf_{\mathrm{cl}!} \epsilon^*$$

is an isomorphism.

Proof. By construction it is enough to treat the case where f is proper. The two functors under consideration are triangulated and way-out. Thus one reduces to the case where the complex is a torsion sheaf placed in degree zero. The result is then the comparison theorem of Exposé XVI. □

Variants 5.3.4. If $\mathcal{A}$ is a sheaf of torsion rings on $S_{\text{ét}}$, the same argument gives an isomorphism between functors

$$D(X, f^*\mathcal{A}) \longrightarrow D(S, \epsilon^*\mathcal{A}).$$

In low degree one also obtains the usual variants for pointed sheaves and for sheaves of ind-finite groups.

Corollary 5.3.5. Let X be a separated scheme of finite type over $\mathbb{C}$, admitting a compactification by a proper $\mathbb{C}$-scheme, and let F be a torsion sheaf on X. Then

$$H^q_!(X, F) \simeq H^q_c(X_{\text{cl}}, \epsilon^* F).$$

Theorem 5.3.6 (finiteness). Let $f : X \to S$ be a compactifiable morphism of finite presentation and let A be a left noetherian torsion ring. The functor

$$Rf_! : D(X, A) \longrightarrow D(S, A)$$

sends complexes with constructible cohomology to complexes with constructible cohomology.

Proof. The assertion is local on S, which may be assumed affine. By passage to the limit one reduces to the noetherian case. The complexes with constructible cohomology form a triangulated subcategory, and $Rf_!$ is way-out; it is therefore enough to prove that, for a constructible sheaf $\mathcal{F}$ of A-modules, the sheaves $R^q f_! \mathcal{F}$ are constructible. Choose a compactification $f = pj$. Then

$$R^q f_! \mathcal{F} = R^q p_*(j_! \mathcal{F}),$$

and $j_!\mathcal{F}$ is constructible. The assertion follows from the finiteness theorem for proper direct images. $\square$

Corollary 5.3.8. Under the same hypotheses, if S is noetherian and K is a bounded complex of constructible A-modules on X, then the formation of $Rf_!K$ is compatible with restriction to locally closed strata and with passage to geometric fibres; in particular the ranks and finite stalk modules occurring in the cohomology sheaves of $Rf_!K$ vary constructibly on S.

This is the usual form in which the finiteness theorem is used below: after compactification the assertion is reduced to the proper case, and all local statements are checked after applying the base-change theorem.

5.4. The Künneth formula

Let

$$\begin{tikzcd} X' \arrow[d, "f"] & Z' \arrow[l, "p"'] \arrow[r, "q"] \arrow[d, "h"] & Y' \arrow[d, "g"] \\ X & Z \arrow[l] \arrow[r] & Y \end{tikzcd}$$

be a diagram over a scheme S, with the lower objects over S, and let $\mathcal{A}$ be a sheaf of torsion rings on S. In the proper case, for complexes K on X' and L on Y', one has the usual Künneth morphism

$$Rf_*K \overset{\mathbf{L}}{\boxtimes}_{\mathcal{A}} Rg_*L \longrightarrow Rh_*(K \overset{\mathbf{L}}{\boxtimes}_{\mathcal{A}} L).$$

When f and g are merely compactifiable, factor them into open immersions followed by proper morphisms. With $Z' = X' \times_S Y'$, extension by zero gives the canonical identification

$$j_{3!}(K \boxtimes_{\mathcal{A}} L) \xrightarrow{\sim} j_{1!}K \boxtimes_{\mathcal{A}} j_{2!}L.$$

Composing the inverse of this isomorphism with the proper Künneth morphism yields a Künneth morphism with proper supports

$$Rf_!K \overset{\mathbf{L}}{\boxtimes}_{\mathcal{A}} Rg_!L \longrightarrow Rh_!(K \overset{\mathbf{L}}{\boxtimes}_{\mathcal{A}} L),$$

for $K \in D^-(X', \mathcal{A})$, $L \in D^-(Y', \mathcal{A})$. The construction is independent of the chosen compactifications.

Theorem 5.4.3. Under the above hypotheses, the Künneth morphism for cohomology with proper supports is an isomorphism.

Proof. By construction it is enough to prove the theorem when the morphisms are proper. Consider a base change of the entire diagram

$$(S_1, X_1, Y_1, Z_1, X'_1, Y'_1, Z'_1) \longrightarrow (S, X, Y, Z, X', Y', Z').$$

The base-change morphisms for $Rf_!$, $Rg_!$, and $Rh_!$ fit into a commutative square with the two Künneth morphisms:

$$\begin{tikzcd} C^*(Rf_!K \overset{\mathbf{L}}{\boxtimes} Rg_!L) \arrow[r, "\sim"] \arrow[d] & C^*Rf_!K \overset{\mathbf{L}}{\boxtimes} C^*Rg_!L \arrow[r, "\sim"] & Rf_{1!}C^*K \overset{\mathbf{L}}{\boxtimes} Rg_{1!}C^*L \arrow[d] \\ C^*Rh_!(K \overset{\mathbf{L}}{\boxtimes} L) \arrow[r, "\sim"] & Rh_{1!}C^*(K \overset{\mathbf{L}}{\boxtimes} L) \arrow[r, "\sim"] & Rh_{1!}(C^*K \overset{\mathbf{L}}{\boxtimes} C^*L). \end{tikzcd}$$

Hence it suffices to check the assertion after arbitrary base change to geometric points of Z. We may therefore assume S is the spectrum of an algebraically closed field and that the structural maps from X and Y to S are the relevant maps. The problem becomes the familiar map

$$Rf_!K \overset{\mathbf{L}}{\otimes}_{\mathcal{A}} Rg_!L \longrightarrow R(f \times g)_!(K \overset{\mathbf{L}}{\boxtimes}_{\mathcal{A}} L).$$

In the proper case the proof is the standard one: apply the projection formula and proper base change to the square defining $X \times_S Y$. The compatibility square shows that the reduction made above is legitimate, and the theorem follows. □

More generally, if $(f_i : X'_i \to X_i)_{i \in I}$ is a finite family of compactifiable morphisms of S-schemes, if $X = \prod_S X_i$, $X' = \prod_S X'_i$, and $f : X' \to X$ is the product, then one has a canonical isomorphism

$$\overset{\mathbf{L}}{\boxtimes}_{i \in I} Rf_{i!}(K_i) \xrightarrow{\sim} Rf_!\Big(\overset{\mathbf{L}}{\boxtimes}_{i \in I} K_i\Big).$$

Neither the definition of this isomorphism nor the definition of its two sides requires the choice of a total order on I.

5.5. Cohomology of symmetric powers

The main result of this section is Theorem 5.5.21. It computes, for suitable coefficients, the cohomology with proper supports of a symmetric power of a scheme in terms of the cohomology with proper supports of the original scheme. The topological analogue for semi-simplicial topology is classical: if X_* is a semi-simplicial set, the geometric realization of the n-th symmetric power $X_*^n/\mathfrak{S}_n$ maps canonically and continuously to the n-th symmetric power of the realization of X_*, and this map induces the expected isomorphism on cohomology.

We recall the algebraic ingredients used in the proof. Lazard's theorem says that every flat module over a ring A is a filtered inductive limit of finite free modules. Consequently filtered inductive limits of finite free, respectively finite projective, modules give precisely the flat, respectively flat Mittag–Leffler/projective-type, modules needed here; flatness is stable under extensions; and every short exact sequence of flat modules is a filtered inductive limit of split short exact sequences.

For an A-module M, write

$$TS^n(M) = (M^{\otimes n})^{\mathfrak{S}_n}$$

for symmetric tensors. If B is a commutative A-algebra, then $TS^n(B)$ is an A-algebra, and $\operatorname{Spec} TS^n_A(B)$ is the n-th symmetric power of $\operatorname{Spec} B$ over $\operatorname{Spec} A$.

Following Roby, a polynomial law $f : M \to N$ is a system of maps

$$f_{A'} : M \otimes_A A' \longrightarrow N \otimes_A A'$$

functorial in the commutative A-algebra A'. It is homogeneous of degree n if $f_{A'}(a'x) = a'^n f_{A'}(x)$. The functor

$$N \longmapsto \operatorname{Hom}^{n\text{-ic}}(M, N)$$

is represented by a module $\Gamma^n(M)$ with universal degree-n polynomial map $\gamma^n : M \to \Gamma^n(M)$. Similarly, multihomogeneous polynomial laws of multidegree $(n_i)_{i\in I}$ are represented by $\bigotimes_i \Gamma^{n_i}(M_i)$.

The functors TS^n and Γ^n commute with filtered inductive limits and carry finite free modules to finite free modules. Thus they preserve flat modules. The canonical degree-n map

$$m \longmapsto m \otimes \cdots \otimes m$$

from M to $TS^n(M)$ defines a natural map

$$\Gamma^n(M) \longrightarrow TS^n(M),$$

which is an isomorphism for finite free M, and hence also for flat M.

If $P_1 \to P_0 \to M \to 0$ is a presentation and $P_1' = P_1 \times P_0$, with maps $j_0, j_1 : P_1' \to P_0$, the sequence

$$\Gamma^n(P_1') \rightrightarrows \Gamma^n(P_0) \longrightarrow \Gamma^n(M) \longrightarrow 0$$

is exact. For a flat presentation this gives the description

$$TS^n(P_1') \rightrightarrows TS^n(P_0) \longrightarrow \Gamma^n(M) \longrightarrow 0.$$

If $M = M' \oplus M''$, there is a canonical decomposition

$$\Gamma^n(M) \simeq \bigoplus_{n'+n''=n} \Gamma^{n'}(M') \otimes \Gamma^{n''}(M'').$$

For a short exact sequence $0 \to M' \to M \to M'' \to 0$, the module $\Gamma^n(M)$ has a canonical decreasing filtration whose graded pieces are

$$\Gamma^i(M'') \otimes \Gamma^{n-i}(M'),$$

provided the sequence is split; by Lazard's theorem the same holds when M'' is flat. The functor Γ^n commutes with extension of scalars.

Dold–Puppe theory identifies co-semi-simplicial objects of an abelian category with non-negatively graded complexes through the normalized-chain functor N. This equivalence and its inverse preserve homotopies. Applying Γ^n levelwise to a co-semi-simplicial resolution and normalizing defines the derived functors $L\Gamma^n$ used below. If a property is stable under finite sums and direct factors, it holds for the semi-simplicial terms up to degree r if and only if it holds for the normalized complex up to degree r.

Lemma 5.5.4. For an exact sequence of complexes satisfying the flatness conditions needed to apply Dold–Puppe, the filtrations on Γ^n associated with the short exact sequence pass to the derived functors $L\Gamma^n$. In particular the associated graded pieces are the sums of the terms

$$L\Gamma^i(K'') \overset{\mathbf{L}}{\otimes} L\Gamma^{n-i}(K').$$

Proposition 5.5.6. If $K \to L$ is a quasi-isomorphism between suitable bounded complexes whose terms satisfy the required flatness hypotheses, then $L\Gamma^n(K) \to L\Gamma^n(L)$ is a quasi-isomorphism. The construction is compatible with finite direct sums, extension of scalars, and the canonical product maps.

The proof proceeds by replacing complexes by co-semi-simplicial objects, filtering by degeneracy degree and by the standard filtration attached to a short exact sequence, and then using the exactness properties of Γ^n recalled above. The key point is that the functors under consideration preserve the flat objects used for the resolutions.

Now let $p : X \to S$ be a quasi-projective morphism. For a sheaf F of $\mathcal{A}$-modules on X, define an external divided-power sheaf $\Gamma^n_{\text{ext}}(F)$ on $\mathrm{Sym}^n_S(X)$. Locally on affines, this is the sheaf associated with $\Gamma^n_A(M)$. The construction is functorial in F, commutes with localization, and is compatible with extension by zero for open immersions:

$$\Gamma^n_{\text{ext}}(j_!F) \simeq \mathrm{Sym}^n_S(j)_!\Gamma^n_{\text{ext}}(F).$$

Derived functors $L\Gamma^n_{\text{ext}}$ are obtained by resolving by complexes of flat modules of Tor-dimension ≤ 0. They satisfy the same formal properties as $L\Gamma^n$.

Let $u : X \to Y$ be a compactifiable morphism of S-schemes. Assume that the symmetric powers $\mathrm{Sym}^n_S(X)$, $\mathrm{Sym}^n_S(\bar{X})$, and $\mathrm{Sym}^n_S(Y)$ exist for a compactification $X \xrightarrow{j} \bar{X} \xrightarrow{\bar{u}} Y$. The symmetric Künneth morphism is the composite

$$L\Gamma^n_{\text{ext}}Ru_! \longrightarrow R\,\mathrm{Sym}^n_S(u)_!L\Gamma^n_{\text{ext}},$$

obtained by writing $Ru_! = R\bar{u}_*j_!$, applying the proper symmetric Künneth morphism for $\bar{u}$, then using the displayed compatibility with $j_!$. It is independent of the compactification, is transitive for composites, and is compatible with extension of scalars, ordinary base change, and radicial base change in the forms needed later.

Theorem 5.5.21 (symmetric Künneth formula). Let S be coherent, i.e. quasi-compact and quasi-separated, and ringed by a sheaf of commutative torsion rings $\mathcal{A}$ whose fibres are noetherian. Let every S-scheme be ringed by the inverse image of $\mathcal{A}$. Let $u : X \to Y$ be a quasi-projective morphism of S-schemes. For every integer $n \geq 0$ and every

$$K \in D^{b,\mathrm{Tor}\leq 0}(X, \mathcal{A}),$$

the symmetric Künneth morphism

$$k^n : L\Gamma^n_{\text{ext}}Ru_!K \longrightarrow R\,\mathrm{Sym}^n_S(u)_!L\Gamma^n_{\text{ext}}K$$

is an isomorphism in $D^{b,\mathrm{Tor}\leq 0}(\mathrm{Sym}^n_S(Y), \mathcal{A})$.

The quasi-projectivity hypothesis is used only to ensure that u has a compactification for which the symmetric powers of X, $\bar{X}$, and Y exist. With algebraic spaces it should be replaceable by compactifiability.

Lemma 5.5.22. Theorem 5.5.21 follows from the following special case. For every algebraically closed field k, every smooth projective curve X over k, and every prime ℓ, the maps

$$L\Gamma^n\Big(R\Gamma(X, \mathbb{Z}/\ell)\Big) \longrightarrow R\Gamma(\mathrm{Sym}^n_k(X), \mathbb{Z}/\ell)$$

are isomorphisms for all $n \geq 0$.

Here $R\Gamma$ means global sections, while $L\Gamma^n$ is the derived divided-power functor. The reduction proceeds by devissage. A short exact sequence of bounded complexes of Tor-dimension ≤ 0

$$0 \to K' \to K \to K'' \to 0$$

has the property that, if the symmetric Künneth morphisms are isomorphisms for two of the three complexes, then they are so for the third. The proof uses the filtration on $L\Gamma^n(K)$ with graded pieces $L\Gamma^{n-i}(K') \overset{\mathbf{L}}{\otimes} L\Gamma^i(K'')$, together with induction on n and the ordinary Künneth formula.

The problem is local on S, and any complex is built by successive extensions from complexes supported on affine locally closed pieces. Thus one reduces to the case in which S is affine and u has relative dimension at most one. Checking on geometric points of $\mathrm{Sym}^n_S(Y)$ reduces to $S = \mathrm{Spec}\, k$, with k algebraically closed. A further radicial and finite-support reduction reduces Y to a finite disjoint union of copies of $\mathrm{Spec}\, k$, and by decomposing over the components of Y one reduces to

$$Y = S = \mathrm{Spec}\, k,\ \dim X \leq 1.$$

The coefficient ring is then a noetherian ring A.

Proposition 5.5.23. Let A be a commutative noetherian ring, let $K \in D^-(A)$, and let $n \in \mathbb{Z}$. The following are equivalent:

(i) K has Tor-dimension $\leq n$;

(ii) for every $x \in \mathrm{Spec}\, A$ and every $i < -n$,

$$H^i(K \overset{\mathbf{L}}{\otimes}_A k(x)) = 0.$$

The nontrivial implication is proved by noetherian induction on closed supports. If a module N is supported in a closed subset F, reduce first to the case in which N is an A/I-module for the reduced ideal defining F. Then compare N with its fibre over a maximal point η of F, using the exact sequence

$$0 \to M_1 \to N \to N \otimes k(\eta) \to M_2 \to 0.$$

The modules M_1, M_2 are supported away from η, so the induction hypothesis applies to them, while the fibre term is a sum of copies of $k(\eta)$. This proves the vanishing for N.

Corollary 5.5.24. A morphism $u : K \to L$ in $D^-(A)$ is an isomorphism if, for every $x \in \operatorname{Spec} A$, the induced morphism

$$K \overset{\mathbf{L}}{\otimes}_A k(x) \longrightarrow L \overset{\mathbf{L}}{\otimes}_A k(x)$$

is an isomorphism in $D^-(k(x))$.

Apply Proposition 5.5.23 to the cone of u.

Continuing the reduction of Lemma 5.5.22, compatibility with extension of scalars and Corollary 5.5.24 reduce the coefficient ring to a field of characteristic $\ell > 0$. Shifting complexes and truncating reduce to sheaves placed in degree zero. Such a sheaf is a filtered inductive limit of constructible sheaves. The trace method reduces to sheaves of the form $j_! A_{X'}$ for $j : X' \to X$ quasi-finite, and transitivity replaces X by X'. Thus we reach the case

$$\dim X \leq 1, \qquad K = A_X.$$

After replacing X by its reduction and cutting off the singular finite set, one reduces to either a smooth curve or a finite set. A final devissage of the constant sheaf over connected components gives the smooth connected curve case; the finite connected case is trivial.

Lemma 5.5.26. Theorem 5.5.21 follows from the two following special cases of Lemma 5.5.22:

Case 1. $k = \mathbb{C}$.

Case 2. k has characteristic $p > 0$ and $\ell = p$.

If the characteristic of k is zero, the Lefschetz principle reduces to $k = \mathbb{C}$. If $\operatorname{char} k = p > 0$ and $\ell \neq p$, lift the smooth projective curve X to a curve X' over the Witt vectors $W(k)$. The morphisms

$$f'_n : \operatorname{Sym}^n_{W(k)}(X') \longrightarrow \operatorname{Spec} W(k)$$

are projective and smooth. By the specialization theorem, the sheaves $R^i f'_{n*}(\mathbb{Z}/\ell)$, as well as the terms $L^i\Gamma^n(Rf'_*(\mathbb{Z}/\ell))$, are locally constant on $\operatorname{Spec} W(k)$. Hence the assertion for the special fibre follows from the corresponding assertion for a geometric generic fibre, which is of characteristic zero.

For Case 1, compare the algebraic statement with the topological one by the comparison theorem. A complex smooth projective curve is triangulable. By the same devissage as before, it is enough to prove the topological statement for a closed simplex D of dimension $0, 1$, or 2. Then

$$R\Gamma(D, \mathbb{Z}/\ell) = \mathbb{Z}/\ell$$

in degree zero, and $\operatorname{Sym}^n(D)$ is contractible, so both sides of the symmetric Künneth map are $\mathbb{Z}/\ell$ in degree zero; the induced map on H^0 is the identity.

For Case 2, one compares the $\mathbb{Z}/p$-statement with a coherent statement for $\mathcal{O}$-modules. Let $p : X \to S$ be quasi-projective. A quasi-coherent sheaf on X is regarded as a sheaf of $\mathcal{O}_X$-modules on the ét-site. The functor Γ^n_{ext} exists for quasi-coherent modules and is compatible with affine computations:

$$\Gamma^n_{\mathrm{ext}}(\widetilde{M}) = \widetilde{\Gamma^n_A(M)}.$$

When p is flat, define $D^{b,\mathrm{Tor}_S\leq 0}(X, \mathcal{O}_X)$ to consist of complexes which, as complexes of $p^{-1}\mathcal{O}_S$-modules, have Tor-dimension ≤ 0. The functor Γ^n_{ext} derives to

$$L\Gamma^n_{\mathrm{ext}} : D^{b,\mathrm{Tor}_S\leq 0}(X, \mathcal{O}_X) \to D^{b,\mathrm{Tor}_S\leq 0}(\mathrm{Sym}^n_S(X), \mathcal{O}).$$

For quasi-projective flat $u : X \to Y$ there is a coherent symmetric Künneth morphism

$$L\Gamma^n_{\mathrm{ext}} Ru_* K \longrightarrow R\,\mathrm{Sym}^n_S(u)_* L\Gamma^n_{\mathrm{ext}} K.$$

Proposition 5.5.34. Under these coherent hypotheses, the coherent symmetric Künneth morphism is an isomorphism.

Proof. The question is local on S, so assume S affine. Localizing on $\mathrm{Sym}^n_S(Y)$, one assumes Y affine. Resolving a quasi-coherent complex by a finite alternating Cech complex associated with an affine cover of X reduces by devissage to the case where $K = Rj_* K_1$ for an affine open immersion $j : X_1 \hookrightarrow X$. Transitivity reduces the proof to the affine case for X, Y, S. Then Ru_* and $L\Gamma^n$ are calculated directly on a bounded complex of flat quasi-coherent modules, and the assertion is the affine identity

$$\Gamma^n_{\mathrm{ext}} u_* \mathcal{F} \simeq \mathrm{Sym}^n_S(u)_* \Gamma^n_{\mathrm{ext}} \mathcal{F},$$

which both sides identify with $\widetilde{\Gamma^n_A(F)}$. ∎

In characteristic p, the Artin–Schreier sequence

$$0 \to \mathbb{Z}/p \to \mathcal{O} \xrightarrow{F-1} \mathcal{O} \to 0$$

relates the $\mathbb{Z}/p$-cohomology to coherent cohomology. The Frobenius $F : x \mapsto x^p$ acts p-linearly on $R\Gamma(X, \mathcal{O})$, on $R\Gamma(\mathrm{Sym}^n_k X, \mathcal{O})$, and therefore on $L\Gamma^n(R\Gamma(X, \mathcal{O}))$. The coherent Künneth isomorphism is compatible with F.

Lemma 5.5.36. Let k be a perfect field of characteristic $p > 0$, let $K \in C^{\geq 0}(\mathbb{Z}/p)$ be a finite-dimensional complex of $\mathbb{Z}/p$-vector spaces, and let $L \in C^{\geq 0}(k)$ be a finite-dimensional complex of k-vector spaces with a p-linear endomorphism F. Let $u : K \to L$ be a morphism such that $H^*(Fu) = H^*(u)$, and assume that

$$0 \to H^i(K) \to H^i(L) \xrightarrow{F-1} H^i(L) \to 0$$

is exact for all i. Then the sequences

$$0 \to H^i(\Gamma^n K) \to H^i(\Gamma^n L) \xrightarrow{F-1} H^i(\Gamma^n L) \to 0$$

are exact.

Since $Fu-u$ is null-homotopic, write $Fu-u = dH+Hd$. Perfection of k gives $FH_0-H_0 = H$, so replacing u by the homotopic map $u-(dH_0+H_0d)$ lets us assume $Fu = u$. The functor sending a finite-dimensional k-space with p-linear F to its fixed part $V^S = \{v \mid Fv = v\}$ is exact; the hypothesis says $K \to L^S$ is a quasi-isomorphism. It remains only to note that fixed parts commute with Γ^n. This follows from the standard decomposition below.

Lemma 5.5.37. A finite-dimensional vector space V over a perfect field k, endowed with a p-linear endomorphism F, has a unique decomposition

$$V = V_0 \oplus V_n$$

with $V^S \otimes k \simeq V_0$, $F(V_n) \subset V_n$, and $F|_{V_n}$ nilpotent.

This completes the proof of the symmetric Künneth theorem.

6. The functor $f_!$

We now give a direct construction of the functor $f_!$ appearing in 5.1.11. This construction will then be used to define the trace morphism for flat quasi-finite morphisms of finite presentation.

6.1. A new construction of $f_!$

Let $f : X \to S$ be locally of finite type. A subset $P \subset X$ is said to be proper over S if it is the image of a closed subscheme of X proper over S.

Proposition 6.1.1. Let $f : X \to S$ be locally of finite type and quasi-separated.

(i) A subset of X proper over S is locally closed, and is closed if f is separated.

(ii) If f is separated, the union of two subsets proper over S is proper over S.

(iii) For every S-scheme U, let $\Phi_X(U)$ be the set of subsets of $U \times_S X$ proper over U. The functor Φ_X is a sheaf for the fpqc topology. If f is separated, Φ_X is a sheaf of filtered ordered sets under inclusion.

The first two assertions are immediate. The presheaf of all subsets of $U \times_S X$ is a sheaf for any topology whose coverings are surjective, and Φ_X is a subsheaf by the descent of properness and finite-type closed images.

If $\mathcal{F}$ is an ét-sheaf of pointed sets on X, the support of a section is the complement of the largest open set on which that section is equal to the distinguished section.

Definition 6.1.2. Let $f : X \to S$ be separated and locally of finite type, and let $\mathcal{F}$ be a pointed sheaf on X. Define $f_!\mathcal{F}$ to be the subsheaf of $f_*\mathcal{F}$ whose sections over an ét-scheme $U \to S$ are the sections of $\mathcal{F}$ on $U \times_S X$ with support proper over U.

Proposition 6.1.1 ensures that this is indeed a sheaf.

Lemma 6.1.3. If $f = gh$ is the composite of two separated morphisms locally of finite type, there is a unique isomorphism

$$f_! \simeq g_! h_!$$

compatible with the inclusions into f_* and $g_* h_*$.

When f is an open immersion, this functor is extension by zero. When f is proper, it is f_*. Hence this construction agrees with the earlier construction whenever both are defined.

Proposition 6.1.4. Let $f : X \to S$ be separated and locally of finite type.

(i) The functor $f_!$ is left exact and commutes with filtered inductive limits.

(ii) If f is quasi-finite, then $f_!$ is exact on abelian sheaves.

(iii) For every Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \big\downarrow{\scriptstyle f'} & & \big\downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S, \end{array}$$

there is a base-change isomorphism $g^* f_! \simeq f'_! g'^*$, compatible with the usual base-change morphism $g^* f_* \to f'_* g'^*$.

Proof. For $U \in S_{\text{ét}}$,

$$f_!\mathcal{F}(U) = \varinjlim_{P \in \Phi(U)} \Gamma_P(\mathcal{F}),$$

where the limit is over the filtered system of supports proper over U. If U is quasi-compact and quasi-separated, the functor $\mathcal{F} \mapsto f_!\mathcal{F}(U)$ commutes with finite projective limits and filtered inductive limits, proving (i). Assertion (ii) follows from (iii) by testing exactness on geometric fibres. For (iii) the question is local on S. Write X as the filtered union of open subschemes of finite type over S, reduce to the finite-type case, and use Chow's lemma to find a projective surjection $p^0 : X^0 \to X$ with fp^0 compactifiable. With $X^1 = X^0 \times_X X^0$, the sequence

$$0 \to \mathcal{F} \to p^0_* p^{0*} \mathcal{F} \rightrightarrows p^1_* p^{1*} \mathcal{F}$$

is exact. Functoriality and left exactness reduce the assertion to sheaves of the form $p^0_* \mathcal{G}$, and then to the compactifiable case, already proved in 5.2.7. $\square$

Counterexample 6.1.6. In general $Rf_!$ is not the derived functor of $f_!$. If X is affine over an algebraically closed field k, then

$$f_!\mathcal{F} = \bigoplus_{x\in X(k)} H^0_x(\mathcal{F}),$$

so the derived functors of $f_!$ are

$$\bigoplus_{x\in X(k)} H^i_x(\mathcal{F}).$$

But if X is smooth, irreducible, and one-dimensional, then for ℓ prime to $\operatorname{char} k$

$$H^2_c(X, \mu_{\ell^n}) \simeq \mathbb{Z}/\ell^n\mathbb{Z}.$$

6.2. The trace morphism in relative dimension zero

The main result is the trace theorem 6.2.3. The technical preliminaries 6.2.1 and 6.2.2 allow the proof to reduce to finite morphisms; this reduction is also essentially visible from the local structure theorem for quasi-finite morphisms.

Let $f : X \to S$ be quasi-finite and separated. For $U \in S_{\text{ét}}$, define $\Psi_f(U)$ to be the category of decompositions of f^*U into open and closed pieces

$$f^*U = \coprod_{i\in I} V_i,$$

where I is a finite pointed set with base point 0, and V_i is finite over U for $i \neq 0$. A morphism from $(V_i)_{i\in I}$ to $(V'_j)_{j\in I'}$ is a map $\sigma : I' \to I$ of pointed sets such that

$$V_i = \bigcup_{j\in\sigma^{-1}(i)} V'_j.$$

We say that φ' refines φ if such a morphism exists.

For an abelian sheaf $\mathcal{F}$ on S, a decomposition $\varphi = (V_i)$ gives maps

$$\tau_\varphi : \mathcal{F}^{I^*}|_U \longrightarrow f_!f^*\mathcal{F}|_U, \qquad I^* = I \setminus \{0\},$$

by summing the adjunction maps along the finite components. If φ' refines φ, the transpose map ${}^t\sigma : \mathcal{F}^{I^*} \to \mathcal{F}^{I'^*}$ makes the corresponding triangle commute.

Proposition 6.2.2.

(i) The fibred category Ψ_f is locally filtered to the right over $S_{\text{ét}}$.

(ii) The maps τ_φ induce an isomorphism

$$\varinjlim_{\varphi \in \Psi_f(U)} \mathcal{F}^{I^*}|_U \xrightarrow{\sim} f_! f^* \mathcal{F}|_U.$$

If S is strictly local with closed point s, the scheme X admits a decomposition in $\Psi_f(S)$ with one component empty over s and all other components finite over S with one-point special fibre. This is final in $\Psi_f(S)$, and the assertion is immediate after taking the stalk at s. The general case follows by passage to strict localizations at geometric points.

Theorem 6.2.3. There is a unique rule which assigns to every separated, flat, quasi-finite morphism of finite presentation $f : X \to S$, and every abelian sheaf $\mathcal{F}$ on S, a trace morphism

$$\mathrm{Tr}_f : f_! f^* \mathcal{F} \longrightarrow \mathcal{F},$$

satisfying:

(Var 1) functoriality in $\mathcal{F}$;

(Var 2) compatibility with every base change;

(Var 3) compatibility with composition of such morphisms;

(Var 4) normalization: if f is finite locally free of constant rank n, the composite

$$\mathcal{F} \to f_* f^* \mathcal{F} = f_! f^* \mathcal{F} \xrightarrow{\mathrm{Tr}_f} \mathcal{F}$$

is multiplication by n.

The properties imply additivity on disjoint unions: if $X = X' \amalg X''$, then Tr_f is the sum of $\mathrm{Tr}_{f'}$ and $\mathrm{Tr}_{f''}$. For the construction, restrict in Ψ_f to decompositions for which each finite component is locally free of constant rank. This subcategory is cofinal. Giving a map $f_! f^* \mathcal{F} \to \mathcal{G}$ is the same as giving compatible maps $\mathcal{F}^{I^*} \to \mathcal{G}$ for all such decompositions. For the trace take the weighted sum

$$(s_i)_{i \in I^*} \longmapsto \sum_{i \in I^*} n_i s_i,$$

where n_i is the rank of $V_i \to U$. Compatibility under refinement is exactly the additivity of ranks. This gives existence. Uniqueness follows from the cofinality and the normalization condition.

There is also a local numerical description. If f is as above and $s \in S$, for a point x over s let $\mathcal{O}_s$ be the strict henselization at s, choose the corresponding local component $\mathcal{O}_x$, and let $n(x)$ be the rank over the generic point. The function $x \mapsto n(x)$ is a weighting of the fibres; the trace is the weighted sum.

The trace produces covariance of cohomology with proper supports for quasi-finite flat morphisms. If

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

with u separated, flat, quasi-finite, and of finite presentation, then for a sheaf $\mathcal{F}$ on Y

$$u_* : f_! u^* \mathcal{F} \simeq g_! u_! u^* \mathcal{F} \xrightarrow{g_!(\mathrm{Tr}_u)} g_! \mathcal{F}.$$

If g is compactifiable and K is a suitable torsion complex, this gives

$$u_* : Rf_! u^* K \longrightarrow Rg_! K,$$

and in the case $S = \operatorname{Spec} k$, k algebraically closed,

$$u_* : H_c^q(X, u^* K) \longrightarrow H_c^q(Y, K).$$

For an étale surjective map $u : X \to Y$, let X_n be the $(n+1)$-fold fibre product of X over Y. The simplicial augmentation $(X_n) \to Y$ gives, for a sheaf $\mathcal{F}$ on Y, augmented sheaves $\mathcal{F}_n = u_{n!} u_n^* \mathcal{F}$.

Proposition 6.2.9. If $u : X \to Y$ is étale, separated, of finite type, and surjective, then for every abelian sheaf $\mathcal{F}$ on Y:

- (i) the sheaves $\mathcal{F}_n$ form a left resolution of $\mathcal{F}$;
- (ii) the complex of nondegenerate chains of this simplicial sheaf is a left resolution of $\mathcal{F}$. If the geometric fibres of u have at most d points, this resolution has length at most d.

By base change it suffices to check the statement over a separably closed field, where it is the homology of a simplex.

For an open covering $(U_i)_{i \in I}$ of Y, ordered when finite, this construction recovers the alternating Cech resolution

$$(\mathcal{F}_n)_{\mathrm{ndeg}} = \bigoplus_{|P|=n+1} j_{P!} j_P^* \mathcal{F},$$

where $U_P = \cap_{i \in P} U_i$.

If $K \in D^-(Y, \mathcal{A})$, these resolutions give the extraordinary Cech resolution of K, and, in a triangle over S with g compactifiable, a spectral sequence

$$E_1^{-p,q} = R^q f_{p!}(u_p^* K) \Longrightarrow R^{-p+q} g_! K.$$

Equivalently,

$$E_2^{-p,q} = \check{H}^p(R^q f_{p!}(u_p^* K)) \Longrightarrow R^{-p+q} g_! K.$$

These are the localization spectral sequences “above.”

Proposition 6.2.11. If f is étale, separated, and of finite type, then $f_!$ is left adjoint to f^*; the trace morphism is the adjunction counit.

For an object X of a site, the restriction functor to X has a left adjoint; the latter is generated by the presheaf

$$V \longmapsto \bigoplus_{\xi \in \mathrm{Hom}(V,X)} F(\xi).$$

In the étale case this left adjoint commutes with base change. Comparing it with the trace-defined $f_!$ on geometric fibres proves the assertion. This interpretation is specific to the étale site and should not be transplanted uncritically to the fpqf site.

6.3. Variant of the trace for continuous coefficients

This subsection proves assertions of Grothendieck concerning continuous coefficients and applies them to the construction of trace morphisms for suitable fppf sheaves.

Let A be a commutative ring. If L and M are finite projective A-modules, the map $u \mapsto \bigwedge^m u$ from $\mathrm{Hom}(L, M)$ to $\mathrm{Hom}(\bigwedge^m L, \bigwedge^m M)$ is polynomial of degree m. The associated construction is functorial and behaves well under base change. It is used to define maps on symmetric powers and norm-like maps attached to finite locally free morphisms.

For a finite locally free morphism $f : X \to T$ of rank n, there is a canonical section

$$t : T \longrightarrow \mathrm{Sym}^n_T(X)$$

which, on affine pieces, is defined by the characteristic polynomial or, equivalently, by the elementary symmetric functions of the finite locally free algebra. The construction is compatible with base change, with addition of finite locally free subschemes, and with composition.

Let G be a commutative group over S, represented in a setting in which torsors and symmetric powers behave well. If K is a G_X-torsor on X, one constructs a G-torsor K_n on $\mathrm{Sym}^n_T(X)$. It is obtained by descent from the external sum of the pullbacks of K to X^n, using the symmetric group action. The construction is compatible with base change, change of group, addition of torsors, addition of the integer n, and multiplication of the integer n. In formulas,

$$(g'_n)^* K_n \simeq (g'^* K)_n, \qquad r(K_n) \simeq r(K)_n,$$
$$(K' + K'')_n \simeq K'_n + K''_n,$$

and if $n = n_1 + n_2$, then for the canonical map

$$\alpha : \mathrm{Sym}^{n_1}_T(X) \times_T \mathrm{Sym}^{n_2}_T(X) \to \mathrm{Sym}^n_T(X)$$

one has

$$\alpha^* K_n \simeq \mathrm{pr}_1^* K_{n_1} + \mathrm{pr}_2^* K_{n_2}.$$

If $\alpha : \mathrm{Sym}^n_T(\mathrm{Sym}^m_T(X)) \to \mathrm{Sym}^{nm}_T(X)$ is the canonical multiplication map, then

$$\alpha^*(K_{nm}) \simeq (K_m)_n.$$

When $f : X \to T$ is finite locally free of rank n, the trace of a torsor is defined by

$$\mathrm{Tr}_f(K) = t^*(K_n),$$

where $t : T \to \mathrm{Sym}^n_T(X)$ is the canonical section. For a morphism locally free only locally on T, the same formula defines $\mathrm{Tr}_f(K)$ locally and the preceding compatibilities solve the descent problem. Thus Tr_f is compatible with base change, with change of group, with addition of torsors, and with composition:

$$\mathrm{Tr}_{fg}(K) \simeq \mathrm{Tr}_f(\mathrm{Tr}_g(K)).$$

For relative curves one obtains the useful divisor form. If $X \to S$ is a smooth quasi-projective curve, D_1, D_2 are relative divisors finite over S, and K is a G-torsor on X, then

$$\mathrm{Tr}_{D_1+D_2}(K) \simeq \mathrm{Tr}_{D_1}(K) + \mathrm{Tr}_{D_2}(K).$$

If

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "p"'] & & Y \arrow[dl, "q"] \\ & S & \end{tikzcd}$$

with X, Y smooth quasi-projective curves over S and u quasi-finite, then for a divisor D on X finite over S

$$\mathrm{Tr}_{D/S}(u^*K) \simeq \mathrm{Tr}_{u_*D/S}(K),$$

and if u is finite, for a divisor E on Y finite over S,

$$\mathrm{Tr}_{E/S}(\mathrm{Tr}_{X/Y} K) \simeq \mathrm{Tr}_{u^*E/S}(K).$$

Proposition 6.3.28. Let A and B be commutative R-algebras and let $u : A \to B$ make B a finite locally free faithful A-module. Then the co-semi-simplicial complex of algebras

$$\left(TS^n_R(B^{\otimes_A(k+1)})\right)_{k\geq 0}$$

is, as a differential co-semi-simplicial $TS^n_R(A)$-module, a split resolution of $TS^n_R(A)$.

The associated complex depends only on B as an A-module with the map $A \to B$. Since $B = A \oplus L$ as an A-module after choosing a splitting locally, one writes an explicit homotopy operator.

Proposition 6.3.29. Let A be a ring, B^* a co-semi-simplicial A-algebra, and $A \to B^*$

the structural map. For extension of scalars to be faithful, respectively fully faithful, it is enough that
$$0 \to A \to B^0,$$
respectively
$$0 \to A \to B^0 \xrightarrow{j_0 - j_1} B^1,$$
be exact and remain exact after tensoring with any A-module.

From Propositions 6.3.28 and 6.3.29 one obtains that the required descent condition holds for every affine group G. More generally it should extend to groups represented by algebraic spaces; in particular one expects the flat algebraic-space case to satisfy the stronger condition.

Proposition 6.3.31. Let
$$0 \to G \xrightarrow{r} G' \xrightarrow{u} H \to 0$$
be an exact sequence over S. Suppose that G' and H, hence G, satisfy the basic representability condition, that G' satisfies the descent condition for the construction of K_n, and that G satisfies the first descent condition. Then G satisfies the second descent condition.

The proof is local for the étale topology on the source. It is enough to treat a G-torsor K for which the induced G'-torsor is trivial. Such a torsor is equivalently a trivial G'-torsor K' together with an isomorphism $H \simeq u(K')$. On $Z = \mathrm{Sym}^n_T(X)$, the induced isomorphism $H_Z \simeq u(K'_n)$ provides precisely the descent datum required for the construction.

7. Appendix

The appendix uses the notation and results of Exposé VI_B. Let $\mathrm{Sch}_!$ be the category whose objects are quasi-compact quasi-separated schemes and whose morphisms are separated morphisms of finite type. Let $\mathcal{E}$ be the bifibred category over $\mathrm{Sch}_!$, with fibres opposite to the corresponding ét-topoi. Fix an object R of $\mathrm{Sch}_!$, and let $\mathcal{E}_R$ be the induced bifibred category over $\mathrm{Sch}_{!R}$. Let $\mathcal{A}$ be a ring object of the topos of sections $\Gamma(\mathcal{E}_R^\circ)$, satisfying:

(i) $\mathcal{A}$ is a cocartesian section over $\mathrm{Sch}_{!R}^\circ$;

(ii) for every $X \in \mathrm{Sch}_{!R}$, $\mathcal{A}_X$ is a torsion sheaf of rings.

The goal is to define, for every morphism $f : X \to Y$ in $\mathrm{Sch}_{!R}$, a functor
$$Rf_! : D(X, \mathcal{A}_X) \longrightarrow D(Y, \mathcal{A}_Y),$$
which coincides with the previously defined functor when f is compactifiable and which makes the categories $D(X, \mathcal{A}_X)$ into a cofibre category over $\mathrm{Sch}_{!R}$. In modern terminology, Nagata compactification makes every morphism of $\mathrm{Sch}_!$ compactifiable; the appendix shows how to avoid using that theorem by relying instead on Chow's lemma.

7.0. Preliminaries

Lemma 7.0.3. Let

$$\begin{tikzcd} X' \arrow[r, "j'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "j"'] & Y \end{tikzcd}$$

be a commutative diagram of schemes, where f, f' are proper and j, j' are immersions. For every complex $K^\bullet \in D^+_{\mathrm{tors}}(X')$, the canonical morphism

$$j_! R^+ f'_* K^\bullet \longrightarrow R^+ f_* j_! K^\bullet$$

is an isomorphism.

Proof. By the lemma on way-out functors, it is enough to treat torsion sheaves in fixed cohomological degree:

$$j_! R^i f'_* F \longrightarrow R^i f_* j_! F.$$

Taking the closed images of j and j', reduce to the case where the immersions are dense open immersions. Properness then makes the square Cartesian. The proper base-change theorem, applied at every geometric point of Y, gives the result. $\square$

Construction 7.0.4. Let D be a category and let X, Y be D-objects of $\mathrm{Sch}_{!R}$ such that, for every arrow $\alpha \to \beta$ of D, the transition maps $X_\beta \to X_\alpha$ and $Y_\beta \to Y_\alpha$ are proper. Let $j : X \to Y$ be a morphism with each $j_\alpha : X_\alpha \to Y_\alpha$ an immersion. The isomorphism of Lemma 7.0.3 for $i = 0$, together with its compatibilities, defines an exact functor

$$j_! : \mathrm{Mod}(\Gamma(\bar{X}), \mathcal{A}) \longrightarrow \mathrm{Mod}(\Gamma(\bar{Y}), \mathcal{A}),$$

and hence a triangulated functor on derived categories. For every object α of D, the square comparing this $j_!$ with the usual extension by zero $j_{\alpha !}$ on the fibre at α is commutative.

Construction 7.0.5. Given a square

$$\begin{tikzcd} X' \arrow[r, "i'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "i"'] & Y \end{tikzcd}$$

with i, i' immersions, f proper, and f' proper and surjective, one obtains a square of simplicial objects

$$\begin{tikzcd} [X'|Y'] \arrow[r, "j"] \arrow[d] & [X|Y] \arrow[d] \\ C^{\Delta}_{Y'} \arrow[r, "i"'] & C^{\Delta}_{Y}, \end{tikzcd}$$

where every j_n is an immersion. There is a canonical isomorphism

$$i_! \simeq R^+\bar{u}_{f*}\, j_!\, L^+\bar{u}^*_{f'}.$$

After replacing by closed images, one may suppose the immersions are dense opens. The square is then Cartesian and f is surjective. The base-change relation gives $L^+\bar{u}^*_f i_! \simeq j_! L^+\bar{u}^*_{f'}$; cohomological descent for u_f gives the displayed isomorphism.

Lemma 7.0.6 (Chow-type lemma). Let S be quasi-compact and quasi-separated and let $f : X \to S$ be separated of finite type. Then there exists a commutative diagram

$$\begin{tikzcd} X' \arrow[rr, "p"] \arrow[dr, "f'"'] & & X \arrow[dl, "f"] \\ & S & \end{tikzcd}$$

with p proper and surjective, and with f' quasi-projective. If f is already proper, p may be chosen projective; if f is an immersion one may impose the evident compatibility with the immersion. The construction is stable under the refinements required in the simplicial arguments below.

7.1. Definition of $R^+f_!$

Definition 7.1.1. A pseudo-compactification of a morphism $f : X \to Y$ in $\mathrm{Sch}_{!R}$ is a triple $(p, j, \bar{f})$ consisting of a proper surjection $p : X' \to X$, an immersion $j : X' \to \bar{X}$, and a proper morphism $\bar{f} : \bar{X} \to Y$, such that $fp = \bar{f}j$ and $\bar{f}j$ is the compactified form of fp.

Chow's lemma provides pseudo-compactifications. For such a triple, set

$$R^+f_!K = R^+\bar{f}_*\, j_!\, L^+\bar{u}^*_p K,$$

where $\bar{u}_p$ denotes the simplicial augmentation attached to the proper hypercover generated by p. The descent formalism makes this independent of refinements in the sense explained below.

Construction 7.1.3. If $(p, j, \bar{f})$ is a pseudo-compactification of $f : X \to Y$, the above formula defines a functor

$$R^+f_! : D^+(X, \mathcal{A}_X) \longrightarrow D^+(Y, \mathcal{A}_Y).$$

For a second pseudo-compactification dominating the first, there is a canonical comparison isomorphism between the two functors.

Construction 7.1.4. For two pseudo-compactifications of the same morphism f, choose a common refinement obtained by fibre product and a further Chow modification. The comparison isomorphisms through the common refinement are independent of all choices. Hence $R^+f_!$ is intrinsically defined on D^+.

The independence is proved by checking the cocycle condition for a triple of pseudo-compactifications. Lemma 7.0.3 handles extension by zero across immersions, and cohomological descent handles the proper surjections. The key technical lemmas are the commutativity of the comparison maps associated with two successive refinements and their compatibility with the descent augmentations.

Proposition 7.1.6. If f is compactifiable in the ordinary sense, the functor $R^+f_!$ just constructed is canonically isomorphic to the functor $R^+f_!$ of Section 5.

Proposition 7.1.7. For a composite $X \xrightarrow{u} Y \xrightarrow{v} Z$ in $\mathrm{Sch}_{!R}$, there is a canonical transitivity isomorphism

$$R^+(vu)_! \simeq R^+v_!R^+u_!.$$

It is compatible with the corresponding transitivity for a triple of composable morphisms.

Definition 7.1.8. The functor $R^+f_!$ is called the direct image with proper supports on bounded-below complexes.

Construction 7.1.9. Given a Cartesian square in $\mathrm{Sch}_{!R}$

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"'] & S, \end{tikzcd}$$

there is a base-change morphism

$$g^*R^+f_!K \longrightarrow R^+f'_!g'^*K.$$

It is defined by choosing compatible pseudo-compactifications of f and f', applying the base-change morphisms already available in the compactifiable case, and checking independence of choices by the comparison theorem for pseudo-compactifications.

7.2. Some properties of $R^+f_!$

Proposition 7.2.1. If the relative dimension of f is at most d, then $R^+f_!$ has cohomological dimension at most $2d$ on torsion sheaves. Equivalently, $R^if_!F = 0$ for $i > 2d$.

This is reduced by pseudo-compactification and cohomological descent to the compactifiable case, where it is Corollary 5.2.8.1.

Proposition 7.2.2. For morphisms in $\mathrm{Sch}_{!R}$, the functor $R^+ f_!$ sends distinguished triangles to distinguished triangles and is compatible with the transitivity isomorphisms. Its formation is local for the ét-topology on the target.

Remark 7.2.3. The construction is intentionally arranged so that all properties are inherited from the compactifiable case after replacing a morphism by a pseudo-compactification. The price paid is the use of cohomological descent in verifying independence of choices.

Proposition 7.2.4. For every Cartesian square as in 7.1.9, the base-change morphism

$$g^* R^+ f_! K \longrightarrow R^+ f'_! g'^* K$$

is an isomorphism.

To prove this, choose pseudo-compactifications compatible with the square. The construction gives a morphism between the two cohomological descent spectral sequences

$$g^* R^q (f \circ \pi_1^p)_! (\pi_1^{p*} K) \Rightarrow g^* R^{p+q} f_! K$$

and

$$R^q (f' \circ \pi_1'^p)_! (\pi_1'^{p*} g'^* K) \Rightarrow R^{p+q} f'_! g'^* K.$$

The E_1-terms are isomorphisms by the compactifiable base-change theorem, hence so is the abutment.

7.3. Definition of $Rf_!$

We now extend $R^+ f_!$ to a reasonable functor

$$Rf_! : D(X, \mathcal{A}_X) \longrightarrow D(Y, \mathcal{A}_Y)$$

for every morphism f of $\mathrm{Sch}_{!R}$.

Choose a pseudo-compactification (p_1, j_1, f_1) of f. For an $\mathcal{A}_X$-module F, take the canonical flasque resolution of

$$j_{1!} L \bar{u}_{p_1}^* F,$$

apply the direct image by f_1, and take the associated simple complex. One obtains a functorial complex

$$0 \to \mathcal{X}^0(F) \xrightarrow{S^1(F)} \mathcal{X}^1(F) \to \cdots \to \mathcal{X}^n(F) \xrightarrow{S^n(F)} \mathcal{X}^{n+1}(F) \to \cdots$$

which represents $R^+ f_!(F)$ in $D^+(Y, \mathcal{A}_Y)$. If the relative dimension of f is at most d, then the cohomology of this complex vanishes in degrees $> 2d$; hence it is canonically represented by the bounded complex

$$0 \to \mathcal{X}^0(F) \to \mathcal{X}^1(F) \to \cdots \to \mathcal{X}^{2d-1}(F) \to \operatorname{Ker} S^{2d}(F) \to 0.$$

Lemma 7.3.2. The functors $\mathcal{X}^0, \mathcal{X}^1, \ldots, \mathcal{X}^{2d-1}$, and $\operatorname{Ker} S^{2d}$, are exact in F.

The exactness of the $\mathcal{X}^n$ follows from exactness of the canonical flasque resolution and from the fact that direct image preserves exact sequences of flasque sheaves. For $\operatorname{Ker} S^{2d}$, apply the long exact homology sequence to the truncated tails

$$\sigma_{\geq 2d+1}\mathcal{X}^\bullet(F).$$

Since the cohomology above degree $2d$ vanishes, the kernel term is exact.

Lemma 7.3.3. Let A and B be abelian categories, and let $J^\bullet$ be a bounded complex of exact functors $A \to B$. The construction which sends a complex $K^\bullet \in C(A)$ to the simple complex associated with the double complex $J^\bullet(K^\bullet)$ defines a triangulated functor $K(A) \to K(B)$ and preserves quasi-isomorphisms.

This follows by writing the construction as

$$C(A) \to C^b(C(B)) \to C(B),$$

where the last arrow is the simple-complex functor, and then applying the standard acyclicity criterion for bounded double complexes with exact rows.

Exercise 7.3.4. For every $K^\bullet \in D^+(X, \mathcal{A}_X)$, the canonical map

$$\tau_{\leq 2d}\mathcal{X}^\bullet(K^\bullet) \to \mathcal{X}^\bullet(K^\bullet)$$

induces a quasi-isomorphism on the associated simple complexes. One observes first that

$$\varinjlim_{k \geq 2d} (\tau_{\leq k}\mathcal{X}^\bullet(K^\bullet))_s \to (\mathcal{X}^\bullet(K^\bullet))_s$$

is an isomorphism, and then uses the fact that cohomology sheaves commute with filtered inductive limits.

Define

$$R^{(1)}f_! : D(X, \mathcal{A}_X) \to D(Y, \mathcal{A}_Y)$$

by the bounded exact-functor complex displayed above. By the preceding exercise, it may be computed directly from the unbounded flasque construction, and it is independent of the pseudo-compactification chosen. The verification amounts to repeating Constructions 7.1.3 and 7.1.4, and Proposition 7.1.6, without the superscript $+$. The original redaction notes that the details are straightforward but lengthy.

Definition 7.3.6. The functor

$$Rf_! : D(X, \mathcal{A}_X) \to D(Y, \mathcal{A}_Y)$$

constructed above is again called the direct-image functor with proper supports.

Final compatibility 7.3.7. The functor $Rf_!$ commutes with base change, and the tensor formula 5.2.9 and the finite Tor-dimension theorem 5.2.10 remain valid for every morphism in $\mathrm{Sch}_{!R}$. These properties justify the preceding construction and put it on the same formal footing as the compactifiable construction of Section 5.

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SGA 4, Exposé XVIII – beginning
The Global Duality Theorem

Opening through the integration formalism for torsors on curves.

Exposé XVIII. Introduction

This exposé is devoted to Poincare duality. The construction follows Verdier's model for separated locally compact topological spaces. One first proves abstractly that the functor $Rf_!$ of Exposé XVII admits a right adjoint, denoted $Rf^!$, and establishes a number of formal properties of this adjoint. Only after this is done does one compute it more explicitly for smooth morphisms.

The possibility of carrying out the construction rests on four properties of $Rf_!$. For the existence of $Rf^!$ and for its formal behavior one uses: first, that $Rf_!$ commutes with base change; second, that the functors $R^i f_!$ commute with filtered inductive limits and vanish for i large; third, that $Rf_!$ can be computed at the level of complexes, that is, from an actual functor between categories of complexes. For the calculation of the adjoint when f is smooth, one uses the computation of $Rf_!((\mathbb{Z}/n\mathbb{Z})_U)$ for sufficiently small U.

Section 3.1, independent of Sections 1 and 2, constructs $Rf^!$. The construction uses special features of the situation, not merely the formal properties just listed. The original redaction explicitly remarks that this makes the section unpleasant and that not every detail of the mechanism was transparent to the redactor. Section 2 treats the calculation of $Rf_!((\mathbb{Z}/n\mathbb{Z})_U)$; it is formal from Section 1, where the cohomology of curves is studied. Section 1 proves more than will later be needed, because it also treats continuous coefficients. It is strongly influenced by Serre.

For later use, the essential points of Section 1 are the trace morphism for curves, treated in 1.1 and generalized in 2.9, and the effacement theorem 1.6.9. A reader willing to accept a transcendental argument may read 1.1 and then go directly to Sections 2 and 3. The main ingredients in 1.6.9 are Poincare duality for smooth curves over an algebraically closed field and the acyclicity lemma used in the proof of smooth base change. The result is generalized in 2.14. Section 3.2 gathers the consequences.

Verdier's method permits the definition of $Rf^!$ for compactifiable morphisms. The price paid for this generality is that the compatibility assertions to be checked are no longer the familiar ones. Some of them, even apparently trivial statements such as 3.2.3, have rather intricate proofs. The appendix to Exposé XVII also allows one to extend the definition to separated morphisms of finite type with coherent target, that is, quasi-compact and quasi-separated target.

1. Cohomology of curves

1.1. The trace morphism

Let $n \geq 1$ be an integer and let X be a scheme on which n is invertible. The sheaf μ_n of n-th roots of unity is a locally free module of rank one over the constant ring sheaf $\mathbb{Z}/n\mathbb{Z}$. For $i \in \mathbb{Z}$ one sets

$$\mathbb{Z}/n\mathbb{Z}(i) = \mu_n^{\otimes i}. \tag{1.1.1.1}$$

If $n = dn'$, exponentiation by d identifies

$$\mu_n \otimes_{\mathbb{Z}/n\mathbb{Z}} \mathbb{Z}/n'\mathbb{Z} \simeq \mu_{n'},$$

and hence gives canonical isomorphisms

$$\mathbb{Z}/n\mathbb{Z}(i) \otimes_{\mathbb{Z}/n\mathbb{Z}} \mathbb{Z}/n'\mathbb{Z} \simeq \mathbb{Z}/n'\mathbb{Z}(i). \tag{1.1.1.2}$$

If F is an abelian sheaf killed by n, put

$$F(i) = F \otimes_{\mathbb{Z}/n\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}(i). \tag{1.1.1.3}$$

This object is independent of the chosen annihilator. For a torsion sheaf prime to the residual characteristics of X, define

$$F(i) = \varinjlim_{m} {}_mF(i), \tag{1.1.1.4}$$

where m runs multiplicatively through integers invertible on X. The functors $F \mapsto F(i)$, the Tate twists, are exact. They commute with inverse image; for quasi-compact and quasi-separated morphisms they also commute with direct image. If $\mathcal{A}$ is a sheaf of rings killed by an integer invertible on X, they extend to exact functors on $D(X, \mathcal{A})$, commuting with the derived inverse and direct image functors.

Definition 1.1.2. A flat curve over a scheme S is a morphism $f : X \to S$ which is flat, separated, of finite presentation, and whose fibres are purely of dimension one.

The phrase "purely of dimension one" does not exclude the empty scheme. In contrast with the convention of EGA, a curve over a field is allowed here to be empty, but it is required to be separated. When S is the spectrum of a field we simply say "curve over S"; such a curve is quasi-projective.

Let now k be an algebraically closed field, let X be a smooth connected curve over k, and let F be a finite abelian group of order prime to $\operatorname{char} k$. The trace map is the canonical isomorphism

$$\mathrm{Tr}_X : H^2_c(X, F(1)) \longrightarrow F.$$

It is defined first for $X = \mathbb{A}^1_k$, using the standard computation of the compactly supported cohomology of the affine line, and then transported to arbitrary smooth curves by excision, normalization, and the compatibility of proper supports with finite maps. For a disconnected curve, the trace is the sum of the traces over the irreducible components.

Lemma 1.1.4. Let X be a smooth curve over an algebraically closed field and let $(U_i)_{i\in c}$ be non-empty opens, one in each irreducible component of X. Then the diagram

$$\begin{tikzcd}
\bigoplus_{i\in c} H^2_c(U_i, F(1)) \arrow[rr, "\sim"] \arrow[dr, "\oplus \mathrm{Tr}_{U_i}"'] & & H^2_c(X, F(1)) \arrow[dl, "\mathrm{Tr}_X"] \\
& F &
\end{tikzcd}$$

is commutative.

Lemma 1.1.5. Let $u : X' \to X$ be a finite flat morphism of smooth curves over an algebraically closed field. The trace maps are compatible with the cohomological trace attached to u:

$$\mathrm{Tr}_{X'} = \mathrm{Tr}_X \circ u_*.$$

The proof reduces to the case where u is finite etale over an open of X. On that open the assertion is a fibrewise summation. The complement is handled by the localization exact sequence for cohomology with proper supports.

Proposition 1.1.6. Let X be a smooth curve over an algebraically closed field and let F be a torsion group prime to the characteristic. The trace map identifies $H^2_c(X, F(1))$ with the group of locally constant functions on the set of connected components of X, with values in F. In particular, for connected X, it is an isomorphism onto F.

This is obtained by applying Lemma 1.1.4 to a dense affine open of each component and using the known calculation for the affine line, then passing to the limit over the prime-to-characteristic torsion.

Lemma 1.1.7. Let $j : U \hookrightarrow X$ be a dense open immersion of smooth curves over an algebraically closed field, with complement Z. The boundary homomorphism in the localization sequence

$$H^1(Z, F) \to H^2_c(U, F(1)) \to H^2_c(X, F(1))$$

is compatible with the local residue maps at the points of Z.

Lemma 1.1.8. Let X be the spectrum of a henselian discrete valuation ring, with generic point η and closed point s, and let F be prime-to-the-residue-characteristic torsion. The boundary map

$$H^1(\eta, F) \to H^0(s, F(-1))$$

is the usual tame residue map.

Lemma 1.1.9. The trace morphism for curves is uniquely characterized by the preceding compatibility with finite flat maps and with the residue maps for henselian traits.

1.2. The one-acyclicity of the affine line

The next goal is to prove an acyclicity assertion for the affine line which will later be used to efface cohomology classes. Let S be a scheme on which n is invertible. Write $p : \mathbb{A}^1_S \to S$ for the projection.

Theorem 1.2.2. For every sheaf F of $\mathbb{Z}/n\mathbb{Z}$-modules on S, the canonical map

$$F \longrightarrow Rp_*p^*F$$

is an isomorphism in degrees 0 and 1, and $R^1p_*p^*F = 0$. Equivalently, $\mathbb{A}^1$ is one-acyclic for prime-to-characteristic torsion coefficients.

The theorem is local on S, and it suffices to prove it after strict henselization at a geometric point. Thus one reduces to the case where S is the spectrum of a separably closed field. For a constant finite sheaf F, the assertion becomes the fact that every finite etale cover of $\mathbb{A}^1$ which is trivial over the geometric generic point in the relevant prime-to-characteristic range is already trivial. The proof is given by a standard compactification argument on $\mathbb{P}^1$ and by the description of tame ramification at infinity.

Lemma 1.2.3. Let S be strictly local and let F be a finite locally constant sheaf on $\mathbb{A}^1_S$ of order invertible on S. If F is trivial on the closed fibre, then it is trivial after a finite etale base change of S; for strictly local S, it is already trivial.

Lemma 1.2.4. Let X be a normal compactification of $\mathbb{A}^1_k$ over an algebraically closed field. A finite etale cover of $\mathbb{A}^1_k$ of prime-to-characteristic degree which is tamely ramified at infinity is controlled by the tame inertia group at infinity. In particular, the prime-to-characteristic part relevant to $R^1p_*p^*F$ vanishes for the affine line.

Combining the two lemmas gives Theorem 1.2.2 for finite locally constant coefficients. General constructible torsion sheaves are then handled by devissage, and arbitrary torsion sheaves by filtered colimits.

The original proof proceeds through a sequence of explicit reductions. First one reduces to affine S, then to S noetherian, and finally to strictly henselian traits and geometric points. The comparison morphism

$$F \longrightarrow p_*p^*F$$

is immediately an isomorphism on stalks. The only point is the vanishing of $R^1p_*p^*F$, which can be tested by finite etale torsors under finite groups. Such torsors extend, after normalization, to covers of $\mathbb{P}^1$ tamely ramified only at infinity, and the tame fundamental group calculation gives the vanishing.

Lemma 1.2.6. Let G be a finite group of order invertible on the base. Every G-torsor on $\mathbb{A}^1_S$ which is trivial on a geometric fibre is locally trivial on S, and over a strictly local base is trivial.

Lemma 1.2.8. The formation of the above vanishing is compatible with passage from a scheme to a limit of affine schemes. Thus the noetherian case implies the general coherent case.

Lemma 1.2.9. The morphism which represents the obstruction to algebraicity of the cohomology class is itself algebraic. Consequently the class that vanishes after every geometric specialization vanishes after an etale localization of the base.

In the language of the original redaction, the last steps verify the algebraicity of a morphism $\widehat{\varphi}$. One first checks the assertion on completed local rings, then descends from formal neighborhoods by finite presentation. This supplies the algebraic torsor whose existence was predicted by the formal calculation.

Variants 1.2.15. The preceding acyclicity remains valid after replacing constant coefficient groups by finite locally constant coefficient groups of order invertible on the base. It is also stable under finite products with a fixed base scheme and under the evident base-change operations.

1.3. Integration of torsors

Let $f : X \to S$ be a smooth projective curve. The integration formalism associates to a torsor on X and to a line bundle, or more generally to a divisor, a torsor on S. This is the relative form of the Weil reciprocity and norm formalism. It will be used to construct pairings and trace maps appearing in duality.

If D is an effective relative divisor finite over S, and if K is a torsor under a commutative group G on X, the trace along D gives a G-torsor

$$\mathrm{Tr}_{D/S}(K)$$

on S. If $\mathcal{L}$ is an invertible sheaf sufficiently ample so that it is represented by such divisors locally on S, one defines a torsor

$$\langle \mathcal{L}, K]$$

by choosing a divisor representing $\mathcal{L}$ and applying the trace. The independence of the divisor follows from the compatibility of trace with rational equivalence.

The construction is additive in the line bundle:

$$\langle \mathcal{L}_1 \otimes \mathcal{L}_2, K] \simeq \langle \mathcal{L}_1, K] + \langle \mathcal{L}_2, K],$$

and additive in the torsor:

$$\langle \mathcal{L}, K_1 + K_2] \simeq \langle \mathcal{L}, K_1] + \langle \mathcal{L}, K_2].$$

It is compatible with every base change $S' \to S$. If $u : S \to X$ is a section and D_u its image, then

$$u^*K = \mathrm{Tr}_{D_u/S}(K) \simeq \langle \mathcal{O}(D_u), K] - \deg(K)\,\mathcal{U},$$

where $\mathcal{U}$ denotes the neutral torsor.

Proposition 1.3.4. The two additivity isomorphisms above are compatible. For line bundles $\mathcal{L}_1, \mathcal{L}_2$ and torsors K_1, K_2, the two ways of identifying

$$\langle \mathcal{L}_1 \otimes \mathcal{L}_2, K_1 + K_2]$$

with the sum of the four torsors

$$\langle \mathcal{L}_i, K_j], \qquad i, j = 1, 2,$$

are equal.

This compatibility, together with the associativity and commutativity constraints, permits one to extend the definition by linearity from sufficiently ample line bundles to arbitrary line bundles.

Assume from now on that the relative curve is smooth and projective and that the group sheaf G satisfies the usual hypotheses guaranteeing effective descent for torsors and compatibility of trace with base change. For a G-torsor K on X, the rule

$$\mathcal{L} \longmapsto \langle \mathcal{L}, K]$$

defines a morphism from the Picard stack of X/S to the stack of G-torsors on S. It is additive, functorial, and compatible with base change.

Formulary 1.3.8. The functor $\langle -, K]$ sends tensor product of line bundles to sum of torsors, sends the trivial line bundle to the neutral torsor, and is compatible with pullback along $S' \to S$. For fixed $\mathcal{L}$, the functor $K \mapsto \langle \mathcal{L}, K]$ is additive in K.

Conversely, given such a morphism of stacks, one recovers a torsor on X by evaluating at the universal point. More precisely, for a section $t : T \to X_T$ over an S-scheme T, the line bundle $\mathcal{O}(t)$ gives a torsor $F(\mathcal{O}(t))$ on T. Taking $T = X$ and t the identity section yields a torsor $\Psi(F)$ on X.

Theorem 1.3.10. Under the hypotheses just stated, the functor

$$\Phi : K \longmapsto \langle -, K]$$

is an equivalence from the category of G-torsors on X to the category of additive, base-change-compatible morphisms from the Picard stack of X/S to the stack of G-torsors on S.

Proof. One reduces to the case where the fibres of X/S are connected and non-empty. Starting from a morphism F, define $\Psi(F)$ by evaluating F on $\mathcal{O}(t)$ for the universal section t. The additivity of F gives canonical identifications showing that $\Phi(\Psi(F)) \simeq F$. Conversely, for a torsor K, the torsor obtained from $\Phi(K)$ by evaluation at the universal point is canonically K. The two identifications are inverse by the compatibility in Proposition 1.3.4. □

Example 1.3.11. For $G = \mathbb{G}_m$, one writes $\langle \mathcal{L}, \mathcal{M} \rangle$ instead of $\langle \mathcal{L}, \mathcal{M}]$. If $\mathcal{L}$ and $\mathcal{M}$ are invertible sheaves on X, then $\langle \mathcal{L}, \mathcal{M} \rangle$ is an invertible sheaf on S. If D is an effective relative divisor, then

$$\langle \mathcal{O}(D), \mathcal{M} \rangle \simeq N_{D/S}(\mathcal{M}).$$

If D and E are disjoint effective relative divisors, this gives a canonical trivialization

$$\sigma_{D,E} : \langle \mathcal{O}(D), \mathcal{O}(E) \rangle \simeq \mathcal{O}.$$

Proposition 1.3.12. Let D_1, D_2, E be effective relative divisors, with D_i disjoint from E. Let f be a rational function with $\operatorname{div}(f) = D_1 - D_2$, equivalently an isomorphism $\mathcal{O}(D_1) \simeq \mathcal{O}(D_2)$. Then the square comparing the trivializations

$$\langle \mathcal{O}(D_i), \mathcal{O}(E) \rangle \simeq \mathcal{O}$$

with multiplication by $N_{E/S}(f)$ is commutative. The analogous square with the two variables interchanged is also commutative.

Corollary 1.3.13 (Weil reciprocity)**.** Let D_1, D_2, E_1, E_2 be effective relative divisors, all intersections required below being empty. If $\operatorname{div}(f) = D_1 - D_2$ and $\operatorname{div}(g) = E_1 - E_2$, then

$$N_{E_1/S}(f) N_{E_2/S}(f)^{-1} = N_{D_1/S}(g) N_{D_2/S}(g)^{-1}.$$

Equivalently,

$$f(\operatorname{div} g) = g(\operatorname{div} f).$$

Corollary 1.3.14. For $G = \mathbb{G}_m$, the degree appearing in the torsor integration formalism is the ordinary degree of an invertible sheaf on the fibres.

The biadditive pairing extends from sufficiently ample line bundles to all invertible sheaves. One obtains the following form.

Formulary 1.3.16. For every smooth projective curve $f : X \to S$, there is a functor

$$(\mathcal{L}, \mathcal{M}) \longmapsto \langle \mathcal{L}, \mathcal{M} \rangle$$

from pairs of invertible sheaves on X to invertible sheaves on S, compatible with arbitrary base change. It is biadditive, symmetric, normalized by $\langle \mathcal{O}, \mathcal{M} \rangle \simeq \mathcal{O}$, and for an effective divisor D satisfies

$$\langle \mathcal{O}(D), \mathcal{M} \rangle \simeq N_{D/S}(\mathcal{M}).$$

It is compatible with rational equivalence of divisors and with the commutativity constraint induced by Weil reciprocity.

For a field k, this pairing admits an explicit expression. Let X be a smooth projective connected curve over k, and let f, g be non-zero rational functions whose divisors have disjoint supports after the usual separation of positive and negative parts. The commutator attached to the pairing is the tame-symbol product

$$\langle f, g \rangle = \prod_x (-1)^{v_x(f)v_x(g)} \left(\frac{f^{v_x(g)}}{g^{v_x(f)}} \right)(x),$$

the product being over the finite set of points where either divisor is supported.

Lemma 1.3.18. With the notation just introduced,

$$\langle f, g \rangle = \prod_x (-1)^{v_x(f)v_x(g)} \left(f^{v_x(g)} / g^{v_x(f)} \right)(x).$$

This is the usual tame-symbol formula. It implies the compatibility assertions of the formulary, in particular the reciprocity relation needed for the symmetry of the pairing.

1.4. Strictly commutative Picard stacks

Some elementary results on Picard stacks are needed in order to express the integration theorem of Section 1.3 as a universal-coefficient formula. The original notes point out that these results are suggested by a 1966 letter of Grothendieck to Verdier, and that the relevant associativity and commutativity conventions for bifunctors are those introduced by Mac Lane.

Let C be a category and let $F : C \times C \to C$ be a functor. An associativity datum on F is a functorial isomorphism

$$\sigma : F(F(X, Y), Z) \xrightarrow{\sim} F(X, F(Y, Z)).$$

It is required to satisfy the usual coherence condition: for every finite family $(X_i)_{i \in I}$, all parenthesized products formed from the X_i by means of F are canonically identified, and the identifications are compatible with further parenthesizing. Equivalently, this is Mac Lane's pentagon condition, although the source formulation uses the free semigroup generated by I.

Suppose in addition that one has functorial isomorphisms

$$\tau_{X,Y} : F(X,Y) \xrightarrow{\sim} F(Y,X).$$

The pair (σ, τ) makes F associative and strictly commutative when the preceding coherence is compatible with passage from the free semigroup to the free abelian semigroup on I. In concrete terms this is the conjunction of the pentagon, the two involutivity conditions for τ, the hexagon, and the extra strict condition that $\tau_{X,X}$ be the identity on $X+X$. The last condition is the point of the word "strictly"; it is not automatic in ordinary symmetric monoidal categories.

Definition 1.4.2. A strictly commutative Picard category $\mathcal{P}$ is a non-empty groupoid, endowed with a functor

$$+ : \mathcal{P} \times \mathcal{P} \to \mathcal{P},$$

associativity and commutativity isomorphisms

$$(X+Y)+Z \simeq X+(Y+Z), \qquad X+Y \simeq Y+X,$$

which make $+$ associative and strictly commutative, and such that for every object X the functor $Y \mapsto X+Y$ is an equivalence of categories.

If $(X_i)_{i \in I}$ is a family of objects in such a category, there is, unique up to unique isomorphism, a functorial way of assigning to every element $\underline{n} \in \mathbb{Z}^{(I)}$ an object $\Sigma(\underline{n})$, with isomorphisms

$$\Sigma(e_i) \simeq X_i, \qquad \Sigma(\underline{n}+\underline{m}) \simeq \Sigma(\underline{n}) + \Sigma(\underline{m}),$$

compatible with all associativity and commutativity constraints. This is the extension from finite sums to formal integer linear combinations.

A strictly commutative Picard category has a unique neutral object up to unique isomorphism. One may define it as an object e together with an isomorphism $e+e \simeq e$. There are then unique unit constraints

$$e+X \simeq X, \qquad X+e \simeq X,$$

compatible with associativity, and the symmetry exchanges them. The group $\mathrm{Aut}(e)$ is abelian, and translation by any object identifies $\mathrm{Aut}(e)$ with the automorphism group of that object.

Definition 1.4.5. A strictly commutative Picard stack $\mathcal{P}$ on a site $\mathcal{S}$ is a stack in groupoids on $\mathcal{S}$, equipped with a sum functor

$$+ : \mathcal{P} \times_{\mathcal{S}} \mathcal{P} \to \mathcal{P}$$

and with associativity and commutativity isomorphisms which, over every object U of the site, make $\mathcal{P}(U)$ a strictly commutative Picard category.

The adjective will usually be omitted below: "Picard stack" always means strictly commutative Picard stack. A Picard stack has a global neutral object e, and $\mathcal{A}ut(e)$ is an abelian sheaf.

An additive functor $F : \mathcal{P}_1 \to \mathcal{P}_2$ between Picard stacks is a morphism of stacks together with a functorial isomorphism

$$F(x+y) \xrightarrow{\sim} F(x) + F(y)$$

compatible with associativity and commutativity. A morphism of additive functors is a morphism of the underlying functors which is compatible with these structural isomorphisms. Such a morphism is automatically an isomorphism.

Given two Picard stacks $\mathcal{P}_1$ and $\mathcal{P}_2$, one obtains another Picard stack

$$\mathcal{HOM}(\mathcal{P}_1, \mathcal{P}_2)$$

whose objects over U are additive functors $\mathcal{P}_1|U \to \mathcal{P}_2|U$, and whose morphisms are the morphisms of additive functors. The sum of two functors is defined objectwise.

Similarly, a biadditive functor $\mathcal{P}_1 \times \mathcal{P}_2 \to \mathcal{P}$ is a functor additive in each variable, with the two additivity structures compatible in the evident square. The biadditive functors form a Picard stack

$$\mathcal{HOM}(\mathcal{P}_1, \mathcal{P}_2; \mathcal{P}).$$

The evaluation map

$$\mathcal{HOM}(\mathcal{P}_1, \mathcal{P}_2) \times \mathcal{P}_1 \to \mathcal{P}_2, \qquad (F, x) \longmapsto F(x),$$

is biadditive. There is also a two-universal biadditive functor

$$\mathcal{P}_1 \times \mathcal{P}_2 \to \mathcal{P}_1 \otimes \mathcal{P}_2,$$

meaning that, for every Picard stack $\mathcal{P}$, composition induces an equivalence

$$\mathcal{HOM}(\mathcal{P}_1 \otimes \mathcal{P}_2, \mathcal{P}) \xrightarrow{\sim} \mathcal{HOM}(\mathcal{P}_1, \mathcal{P}_2; \mathcal{P}). \tag{1.4.8.1}$$

If $u : \mathcal{S}_1 \to \mathcal{S}_2$ is a morphism of sites and $\mathcal{P}$ is a Picard stack on $\mathcal{S}_1$, the direct image $u_*\mathcal{P}$ is the Picard stack on $\mathcal{S}_2$ given by

$$(u_*\mathcal{P})(V) = \mathcal{P}(u^*V).$$

This construction is functorial in the two-categorical sense.

One also needs the stackification process. A Picard prestack on $\mathcal{S}$ is a prestack in groupoids carrying the same Picard operations fibrewise. If $j : \mathcal{P} \to a\mathcal{P}$ is the stack associated with such a prestack, there is a unique Picard-stack structure on $a\mathcal{P}$, up to essentially unique equivalence, for which j is additive. Moreover, for every Picard stack $\mathcal{Q}$, pullback by j gives an equivalence

$$\mathcal{HOM}(a\mathcal{P},\mathcal{Q}) \xrightarrow{\sim} \mathcal{HOM}(\mathcal{P},\mathcal{Q}). \tag{1.4.10.1}$$

The basic algebraic model for Picard stacks is provided by two-term complexes. Let $C^{[-1,0]}(\mathcal{S})$ be the category of complexes of abelian sheaves

$$K: \quad K^{-1} \xrightarrow{d} K^0$$

concentrated in degrees -1 and 0. To such a complex one associates a Picard prestack $\mathrm{pch}(K)$ as follows. Over U, the objects are sections of $K^0(U)$. A morphism from x to y is a section $f \in K^{-1}(U)$ such that $df = y - x$. Composition and addition are the additions in K^{-1} and K^0. Let

$$\mathrm{ch}(K)$$

be the Picard stack associated to this prestack. Then

$$\mathcal{H}^0(K)$$

is the sheaf associated to the presheaf of isomorphism classes of objects of $\mathrm{ch}(K)$, and

$$\mathcal{H}^{-1}(K) \simeq \mathcal{A}ut(e),$$

where e is the neutral object.

A morphism of complexes $f : K \to L$ induces an additive functor

$$\mathrm{ch}(f) : \mathrm{ch}(K) \to \mathrm{ch}(L).$$

The functor $\mathrm{ch}(f)$ is an equivalence if and only if f is a quasi-isomorphism. If $f, g : K \to L$ are two morphisms, then a morphism of additive functors $\mathrm{ch}(f) \to \mathrm{ch}(g)$ is the same thing as a homotopy $H : K \to L[-1]$ satisfying

$$g - f = dH + Hd. \tag{1.4.12.4}$$

Lemma 1.4.13. Every Picard stack is equivalent to $\mathrm{ch}(K)$ for some complex $K \in C^{[-1,0]}(\mathcal{S})$. Moreover, every additive functor

$$F : \mathrm{ch}(K) \to \mathrm{ch}(L)$$

may, after replacing K by a quasi-isomorphic complex K', be represented by a morphism of complexes $K' \to L$.

The proof is explicit. Choose local objects $k_i \in \mathcal{P}(U_i)$ which locally generate the stack. Put $K^0 = \bigoplus_i \mathbb{Z}_{U_i}$. The universal formal sums of the k_i define an additive functor $\mathrm{ch}(0 \to K^0) \to \mathcal{P}$. Define K^{-1} to be the sheaf of pairs consisting of a local section x of K^0 and an isomorphism from the image of 0 to the image of x. The differential records x. This produces a complex K and an equivalence $\mathrm{ch}(K) \simeq \mathcal{P}$. The assertion about additive functors is obtained by choosing local representatives for the image of generators and then pulling back K^{-1} along a local epimorphism onto K^0.

Proposition 1.4.15. The construction ch induces an equivalence of categories

$$D^{[-1,0]}(\mathcal{S}) \xrightarrow{\sim} \mathrm{Ch}^\flat(\mathcal{S}),$$

where $D^{[-1,0]}(\mathcal{S})$ is the full subcategory of the derived category consisting of complexes with cohomology only in degrees -1 and 0, and $\mathrm{Ch}^\flat(\mathcal{S})$ is the category of Picard stacks with morphisms given by isomorphism classes of additive functors.

The inverse equivalence is denoted by $\mathcal{P} \mapsto \mathcal{P}^\flat$. Thus a Picard stack is represented, up to the appropriate equivalence, by its associated two-term derived object.

If $L \in C^{[-1,0]}(\mathcal{S})$ has L^{-1} injective, then $\mathrm{pch}(L)$ is already a stack, and every additive functor $\mathrm{ch}(K) \to \mathrm{ch}(L)$ is isomorphic to $\mathrm{ch}(f)$ for an actual morphism of complexes $f : K \to L$. Consequently the construction ch may be regarded as a two-equivalence between Picard stacks and two-term complexes with injective term in degree -1, with two-morphisms given by homotopies.

Under this dictionary one obtains the following formulas:

$$\mathcal{HOM}(\mathcal{P}_1, \mathcal{P}_2)^\flat \simeq \tau_{\leq 0} R\mathcal{H}om(\mathcal{P}_1^\flat, \mathcal{P}_2^\flat), \tag{1.4.18}$$

$$(f_*\mathcal{P})^\flat \simeq \tau_{\leq 0} Rf_*(\mathcal{P}^\flat), \tag{1.4.19}$$

and

$$(\mathcal{P}_1 \otimes \mathcal{P}_2)^\flat \simeq \tau_{\geq -1}(\mathcal{P}_1^\flat \otimes^{\mathbf{L}} \mathcal{P}_2^\flat). \tag{1.4.20}$$

If G is an abelian sheaf, then $G[1]$, with G placed in degree -1, corresponds to the Picard stack of G-torsors. If $K \in C^{[-1,0]}(\mathcal{S})$, extensions

$$0 \to G[0] \to E \to K \to 0$$

form a Picard stack $\mathcal{EXT}(K, G)$, with addition given by the Baer sum. Sending an extension to the torsor over $x \in K^0$ consisting of lifts of x gives an equivalence

$$\mathcal{EXT}(K, G) \xrightarrow{\sim} \mathcal{HOM}(\mathrm{ch}(K), \mathrm{ch}(G[1])). \tag{1.4.23}$$

Finally, if F is an object of the site and $\mathbb{Z}^{(F)}$ is the abelian sheaf freely generated by the represented sheaf F, then giving an additive functor

$$\mathrm{ch}(\mathbb{Z}^{(F)}) \to C$$

is the same as giving an object of $C(F)$. If $f : \mathcal{S}/F \to \mathcal{S}$ is the canonical morphism of sites,

this gives
$$\mathcal{HOM}(\mathrm{ch}(\mathbb{Z}^{(F)}), C) \cong f_* f^* C. \tag{1.4.24.1}$$
At the level of complexes the corresponding statement is
$$R\mathcal{H}om(\mathbb{Z}^{(F)}, K) \simeq Rf_* f^* K, \tag{1.4.24.2}$$
for $K \in D^+(\mathcal{S})$.

1.5. The universal-coefficients formula

Let $f : X \to S$ be a smooth projective curve. Denote by
$$\mathrm{PIC}(X/S)$$
the Picard stack on the big fppf site of S obtained as the direct image of the stack of invertible sheaves on X. In the notation of the preceding section,
$$\mathrm{PIC}(X/S) = f_* \,\mathrm{ch}(\mathbb{G}_m[1]), \qquad \mathrm{PIC}(X/S)^\flat = \tau_{\leq 0} Rf_*(\mathbb{G}_m[1]). \tag{1.5.1}$$
Every section t of X/S defines the invertible sheaf $\mathcal{O}(t)$. Thus one obtains a morphism of Picard stacks
$$\mathrm{ch}(\mathbb{Z}^{(X)}) \longrightarrow \mathrm{PIC}(X/S). \tag{1.5.1.3}$$
For every Picard stack C on S, this gives a functor
$$\mathcal{HOM}(\mathrm{PIC}(X/S), C) \longrightarrow \mathcal{HOM}(\mathrm{ch}(\mathbb{Z}^{(X)}), C) \simeq f_* f^* C. \tag{1.5.1.4}$$
In derived terms, if $K = C^\flat$, the same map is represented by
$$R\mathcal{H}om(\tau_{\leq 0} Rf_*(\mathbb{G}_m[1]), K) \longrightarrow Rf_* f^* K. \tag{1.5.1.6}$$
For $C = \mathrm{ch}(G[1])$, the theorem of Section 1.3 says exactly that this functor is an equivalence.

Theorem 1.5.2 (universal coefficients)**.** *Let $f : X \to S$ be a smooth projective curve, let C be a Picard stack on S_{fppf}, and put $K = C^\flat$. Assume locally on S that K is represented in $D^+(S_{\mathrm{fppf}})$ by a two-term complex*
$$G_{-1} \to G_0$$
whose terms satisfy the hypotheses imposed in Section 1.3 on coefficient groups. Then:

(I) *The functor*
$$\mathcal{HOM}(\mathrm{PIC}(X/S), C) \longrightarrow f_* f^* C \tag{1.5.2.1}$$
is an equivalence of Picard stacks.

(II) The morphism

$$\tau_{\leq 0} R\mathcal{H}om(\tau_{\leq 0} Rf_*(\mathbb{G}_m[1]), K) \longrightarrow \tau_{\leq 0} Rf_* f^* K \tag{1.5.2.2}$$

is an isomorphism.

The complex $\tau_{\leq 0} Rf_*(\mathbb{G}_m[1])$ therefore plays, for the curve X/S, the role played by homology in the classical universal-coefficient theorem. For a fixed C, the two assertions are equivalent under the correspondence between Picard stacks and two-term complexes. The assertion is local on S. It has already been proved when $C = \mathrm{ch}(G[1])$, by Theorem 1.3.10. For a two-term complex $G_{-1} \to G_0$, one compares the long exact cohomology sequence for Rf_*K with the corresponding sequence for the left hand side of (1.5.2.2), and applies the five lemma.

The formula is functorial for finite flat morphisms of curves. If

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "f_1"'] & & Y \arrow[dl, "f_2"] \\ & S & \end{tikzcd}$$

is flat between smooth projective curves over S, then the norm map on Picard stacks and pullback of sections give an essentially commutative square

$$\begin{tikzcd} \mathbb{Z}^{(X)} \arrow[r] \arrow[d, "u"'] & \mathrm{PIC}(X/S) \arrow[d, "N_{X/Y}"] \\ \mathbb{Z}^{(Y)} \arrow[r] & \mathrm{PIC}(Y/S). \end{tikzcd}$$

For every Picard stack C this yields the expected compatibility of universal coefficients with pullback along u. At the derived level this is the commutativity of the square

$$\begin{tikzcd} R\mathcal{H}om(\tau_{\leq 0} Rf_{2*}\mathbb{G}_m[1], K) \arrow[r] \arrow[d] & Rf_{2*} f_2^* K \arrow[d, "u^*"] \\ R\mathcal{H}om(\tau_{\leq 0} Rf_{1*}\mathbb{G}_m[1], K) \arrow[r] & Rf_{1*} f_1^* K. \end{tikzcd} \tag{1.5.3.3}$$

For torsor coefficients G, the trace for u gives the transposed compatibility: the trace on cohomology corresponds to pullback/norm on the Picard object.

Locally for the etale topology on S, one may split non-canonically

$$\tau_{\leq 0} Rf_* \mathbb{G}_m[1] \simeq f_*\mathbb{G}_m[1] \oplus R^1 f_* \mathbb{G}_m.$$

This gives, for an abelian sheaf G satisfying the standing hypotheses, an isomorphism

$$\mathcal{H}om(R^1 f_* \mathbb{G}_m, G) \xrightarrow{\sim} \mathcal{H}om_S(X, G) \tag{1.5.4.1}$$

and an exact sequence

$$0 \to \mathcal{E}xt^1(R^1 f_* \mathbb{G}_m, G) \to R^1 f_* G \to \mathcal{H}om(f_* \mathbb{G}_m, G) \to 0. \tag{1.5.4.2}$$

The first map and the first term of the exact sequence are induced by the canonical map from X to its relative Picard scheme; the last map is the degree map. After the usual local decomposition $\mathrm{Pic}_{X/S} \simeq \mathrm{Pic}^0_{X/S} \times \mathbb{Z}^r$, the exact sequence may be written

$$0 \to \mathcal{E}xt^1(\mathrm{Pic}^0_{X/S}, G) \to R^1 f_* G \to \mathcal{H}om(f_* \mathbb{G}_m, G) \to 0. \tag{1.5.4.3}$$

Putting $K = \mathbb{G}_m[1]$ in Theorem 1.5.2 gives an autoduality theorem for $\tau_{\leq 0} R f_* \mathbb{G}_m[1]$. A representative complex has a canonical three-step filtration with successive quotients, up to unique quasi-isomorphism,

$$\mathrm{Pic}_{X/S} / \mathrm{Pic}^0_{X/S}, \qquad \mathrm{Pic}^0_{X/S}, \qquad (f_* \mathbb{G}_m)[1].$$

The corresponding dualities are the natural duality between the discrete component group and $f_* \mathbb{G}_m$, and the autoduality of the Jacobian $\mathrm{Pic}^0_{X/S}$.

The non-proper case is more delicate. Let

$$\bar{f} : \bar{X} \to S$$

be proper, of finite presentation, with fibres purely of dimension one; let $Y \subset \bar{X}$ be finite over S, defined by an ideal $\mathfrak{m}$, and let $X = \bar{X} - Y$ be smooth over S. For a torsion etale sheaf G, prime to the residual characteristics, one introduces $\mathbb{G}_m(\mathfrak{m})$, the sheaf of invertible functions on $\bar{X}$ equal to 1 along Y. A $\mathbb{G}_m(\mathfrak{m})$-torsor is the same as an invertible sheaf on $\bar{X}$ rigidified along Y. Relative divisors on X, finite over S, give such objects.

The technical input is that sufficiently positive rigidified line bundles have enough sections. If S is local with closed point s, $\mathcal{O}_{\bar{X}}(1)$ is relatively ample and rigidified along Y, and $\mathcal{M}$ is a rigidified invertible sheaf, then for $n \gg 0$ every section of $\mathcal{M}(n)$ over the fibre $\bar{X}_s$, equal to 1 on Y_s, lifts locally on S to a section equal to 1 on Y. The obstruction lies in $H^1(\bar{X}, \mathcal{N}(n))$ for a coherent sheaf $\mathcal{N}$, and vanishes for n large.

One then imposes the following general-position conditions on sections of a rigidified invertible sheaf $\mathcal{L}$. A section m should be equal to 1 on Y, and its zero scheme $Z(m) \subset X$ should be a relative divisor, hence finite over S. For two or three sections one requires the same condition for the affine pencils

$$\lambda m_1 + (1 - \lambda) m_2, \qquad \lambda m_1 + \mu m_2 + (1 - \lambda - \mu) m_3.$$

By a general-position argument, after passing to an etale neighbourhood of a geometric point of S, finitely many sections may be joined by auxiliary sections satisfying these conditions.

Let now K be a G-torsor on X. If (m_1, m_2) is such a good pair of sections of $\mathcal{L}$, trace along the divisor

$$\lambda m_1 + (1-\lambda) m_2 = 0$$

gives a G-torsor on the affine line over S. By the acyclicity lemma for affine space, this torsor is pulled back from S, and the specializations at 0 and 1 yield a canonical isomorphism

$$\mathrm{Tr}_{Z(m_1)/S}(K) \xrightarrow{\sim} \mathrm{Tr}_{Z(m_2)/S}(K). \tag{1.5.12.1}$$

This isomorphism is independent of the auxiliary choices. If E is a relative divisor on X, then replacing $\mathcal{L}$ by $\mathcal{L} \otimes \mathcal{O}(E)$ changes the trace torsor by adding $\mathrm{Tr}_{E/S}(K)$. Hence one can define, locally and then by descent, a torsor

$$\langle \mathcal{L}, K]$$

which is additive in $\mathcal{L}$.

Automorphisms of $\mathcal{L}$ act trivially on $\langle \mathcal{L}, K]$. Indeed, the sheaf $\bar{f}_*\mathbb{G}_m(\mathfrak{m})$ is connected by affine lines: any two sections are joined, over an open of the affine line with non-empty fibres, by $\lambda x + (1-\lambda)y$. Since every morphism from such an open to the torsion group G is constant, the automorphism action is trivial. Thus $\langle \mathcal{L}, K]$ depends only on the class of $\mathcal{L}$ in $R^1\bar{f}_*\mathbb{G}_m(\mathfrak{m})$.

Consequently, for every section λ of $R^1\bar{f}_*\mathbb{G}_m(\mathfrak{m})$ one obtains a torsor $\langle \lambda, K]$. This construction is additive in λ, hence gives an extension $e(K)$ of $R^1\bar{f}_*\mathbb{G}_m(\mathfrak{m})$ by G. It is compatible with base change and with addition of torsors, so there is an additive functor

$$e : \bar{f}_* \,\mathrm{ch}(G[1]) \longrightarrow \mathcal{E}XT(R^1\bar{f}_*\mathbb{G}_m(\mathfrak{m}), G).$$

Let $j : X \to R^1\bar{f}_*\mathbb{G}_m(\mathfrak{m})$ be the map sending a point t to the class of $\mathcal{O}(t)$. Pullback by j gives the inverse functor.

Proposition 1.5.14. Under the hypotheses above, the functors e and j^* are inverse equivalences of Picard stacks. In particular there is an isomorphism

$$\mathrm{Ext}^1(R^1\bar{f}_*\mathbb{G}_m(\mathfrak{m}), G) \xrightarrow{\sim} H^1(X, G).$$

This identification is compatible with finite flat morphisms of compactified curves. If $\bar{u} : \bar{X}_1 \to \bar{X}_2$ carries Y_1 into the inverse image of Y_2 and induces a finite flat morphism $X_1 \to X_2$, then trace on torsors corresponds to pullback of rigidified line bundles. If $\bar{u}$ itself is finite flat and $Y_1 = \bar{u}^{-1}Y_2$, then pullback of torsors corresponds to the norm of rigidified line bundles.

1.6. An effacement theorem

The present subsection, from Lemma 1.6.6 onward, may be read independently of Sections 1.2 through 1.5.

Lemma 1.6.1. Let X be a scheme, let $Y \subset X$ be a closed subscheme defined by an ideal $\mathfrak{m}$, and put $U = X - Y$, with immersions

$$U \xrightarrow{j} X \xleftarrow{i} Y.$$

Let $\mathbb{G}_m(\mathfrak{m})$ be the fpqc sheaf whose sections over an X-scheme are the invertible functions equal to 1 on the inverse image of Y. For n invertible on X, the sequence

$$0 \to j_!\mu_n \to \mathbb{G}_m(\mathfrak{m}) \xrightarrow{n} \mathbb{G}_m(\mathfrak{m}) \to 0$$

is exact on the big etale site of X.

The proof is the Kummer sequence for $\mathbb{G}_m$ together with the exact sequence relating functions on X, functions on Y, and functions congruent to 1 modulo $\mathfrak{m}$. Since the direct image from the closed subscheme is exact, a diagram with exact columns and the usual Kummer rows gives the assertion.

Let k be a field. The notes recall that, under standard representability hypotheses, a sheaf which occurs as the middle term of an exact sequence of fppf sheaves with representable neighbouring terms is representable and locally of finite type. If P is a connected commutative algebraic group of finite type over k, and n is invertible in k, then

$$0 \to {}_nP \to P \xrightarrow{n} P \to 0 \tag{1.6.2.1}$$

is exact. If G is a finite etale commutative group killed by n, then

$$\mathrm{Hom}({}_nP, G) \xrightarrow{\sim} \mathrm{Ext}^1(P, G). \tag{1.6.2.2}$$

Assume now that X is proper over $\operatorname{Spec} k$. The exact sequence

$$0 \to \mathbb{G}_m(\mathfrak{m}) \to \mathbb{G}_m \to i_*\mathbb{G}_m \to 0$$

gives

$$0 \to p_*\mathbb{G}_m(\mathfrak{m}) \to p_*\mathbb{G}_{m,X} \to (pi)_*\mathbb{G}_{m,Y} \to R^1p_*\mathbb{G}_m(\mathfrak{m}) \to \mathrm{Pic}_{X/k} \to \mathrm{Pic}_{Y/k}\,. \tag{1.6.3.1}$$

The terms are represented by group schemes of the expected finiteness type. Therefore $p_*\mathbb{G}_m(\mathfrak{m})$ and $R^1p_*\mathbb{G}_m(\mathfrak{m})$ are representable.

Definition 1.6.4. The group scheme representing $R^1p_*\mathbb{G}_m(\mathfrak{m})$ is denoted

$$\mathrm{Pic}_{\mathfrak{m},X/k}\,.$$

Its neutral component is $\mathrm{Pic}^0_{\mathfrak{m},X/k}$, and $\mathrm{Pic}^\tau_{\mathfrak{m},X/k}$ is the inverse image in $\mathrm{Pic}_{\mathfrak{m},X/k}$ of the torsion subgroup of the component group.

For algebraically closed k and n invertible in k, the Kummer sequence of Lemma 1.6.1 gives

$$H^1_c(U, \mu_n) \xrightarrow{\sim} {}_n\,\mathrm{Pic}^\tau_{\mathfrak{m},X/k}(k). \tag{1.6.4.1}$$

Let X now be a smooth curve over an algebraically closed field k, let $\bar{X}$ be its smooth projective compactification, and let $Y = \bar{X} - X$. For a finite abelian group G killed by an integer n prime to $\mathrm{char}\, k$, the preceding identifications, together with Proposition 1.5.14 and (1.6.2.2), give a canonical isomorphism

$$\mathrm{Hom}(H^1_c(X, \mu_n), G) \xrightarrow{\sim} H^1(X, G). \tag{1.6.5}$$

If $u : X \to Y$ is finite between smooth curves, this isomorphism is compatible with trace on compactly supported cohomology and pullback on ordinary cohomology.

Lemma 1.6.6. Let X be a smooth curve over an algebraically closed field k, let n be invertible in k, and let $u : X' \to X$ be a tame etale covering. Then

$$\mathrm{Tr}_u : H^1_c(X', \mathbb{Z}/n) \to H^1_c(X, \mathbb{Z}/n)$$

is, non-canonically unless an isomorphism $\mu_n \simeq \mathbb{Z}/n$ has been chosen, the transpose of

$$u^* : H^1(X, \mathbb{Z}/n) \to H^1(X', \mathbb{Z}/n).$$

One proof is immediate from (1.6.5) after identifying μ_n with $\mathbb{Z}/n$. A second proof uses comparison with classical topology over $\mathbb{C}$, where the result is the usual transposition under Poincare duality; the characteristic-zero case follows by Lefschetz principle, and the positive-characteristic case is obtained by lifting the curve and the tame covering to the Witt vectors and applying base change to the locally constant sheaves $R^1 f_* \mathbb{Z}/n$ and $R^1 f_! \mathbb{Z}/n$.

Lemma 1.6.7. Let X be a smooth connected curve over an algebraically closed field k, let p be the characteristic exponent of k, and let n be prime to p. There exists a principal finite etale Galois cover $u : X' \to X$, of degree dividing a power of n, such that

$$\mathrm{Tr}_u : H^1_c(X', \mathbb{Z}/n) \to H^1_c(X, \mathbb{Z}/n)$$

is zero.

Indeed, take the largest connected abelian principal cover of X with Galois group killed by n. By construction, pullback kills $H^1(X, \mathbb{Z}/n)$. Lemma 1.6.6 then says that the trace on compactly supported cohomology is zero.

Lemma 1.6.8. Let $f : X \to S$ be smooth and let $x : S \to X$ be a section. There exists a Zariski open neighbourhood U of $x(S)$ in X such that the geometric fibres of $U \to S$ are connected.

Fundamental Lemma 1.6.9. Let $f : X \to S$ be a compactifiable smooth curve, let x be a geometric point of X, let s be its image in S, and let n be invertible on S. Then there exist an etale neighbourhood V of s in S and an etale neighbourhood U of x in $X_V = X \times_S V$, fitting into

$$\begin{tikzcd} x \arrow[r] \arrow[d] & U \arrow[r, "u"] \arrow[dr, "f'"'] & X_V \arrow[r] \arrow[d, "f_V"] & X \arrow[d, "f"] \\ s \arrow[rr] & & V \arrow[r] & S, \end{tikzcd} \tag{1.6.9.1}$$

such that

$$R^0 f'_!(\mathbb{Z}/n) = 0,$$

the trace map

$$\mathrm{Tr}_u : R^1 f'_!(\mathbb{Z}/n) \to R^1 f_{V!}(\mathbb{Z}/n)$$

is zero, and the curve trace

$$\mathrm{Tr}_{f'} : R^2 f'_!(\mathbb{Z}/n(1)) \to \mathbb{Z}/n$$

is an isomorphism.

For the first condition it is enough to arrange that f' be quasi-affine. For the last it is enough that the geometric fibres of f' be connected; Lemma 1.6.8 supplies this after an etale localization and shrinking. It remains to kill the trace in degree one. The problem reduces to the case of geometrically connected fibres. For a given geometric point t of S, Lemma 1.6.7 supplies, on the fibre X_t, an etale principal cover whose trace on $H^1_c(-, \mathbb{Z}/n)$ is zero. By the acyclicity theorem used in smooth base change, there is an etale neighbourhood $U \to X$ of x such that, over the fibre at t, the map $U_t \to X_t$ factors through that cover. The fibre at t of

$$\mathrm{Tr}_u : R^1 f'_!(\mathbb{Z}/n) \to R^1 f_!(\mathbb{Z}/n) \tag{1.6.9.2}$$

is then zero. As U varies, the images of these maps form a filtered decreasing system of subobjects of the constructible sheaf $R^1 f_!(\mathbb{Z}/n)$, and the stalk is eventually zero at every geometric point. By noetherian induction the system is eventually zero globally. This completes the proof of the fundamental lemma.

2. The trace morphism

This section proves the general trace morphism and the local effacement theorem which will be used in the global duality theorem. The proof of the principal trace theorem uses only the results of Section 1.

Lemma 2.1. Let $f : X \to S$ be compactifiable of relative dimension $\leq d$, and let $j : U \hookrightarrow X$ be an open subscheme of finite type whose complement has fibre dimension $< d$. For every torsion sheaf F on X, the canonical map

$$R^{2d}(fj)_! j^* F \xrightarrow{\sim} R^{2d} f_! F$$

is an isomorphism.

Let $i : Y = X - U \hookrightarrow X$. The localization exact sequence for $f_!$ contains

$$R^{2d-1}(fi)_! i^* F \to R^{2d}(fj)_! j^* F \to R^{2d} f_! F \to R^{2d}(fi)_! i^* F.$$

The two outside terms vanish by the cohomological-dimension estimate, since the fibres of Y have dimension $< d$.

Lemma 2.2. Let $f : X \to S$ be compactifiable of relative dimension $\leq d$. Let $u_i : X_i \to X$ be a family of etale, separated, finite-type morphisms, and write $X_{ij} = X_i \times_X X_j$. Then for every torsion sheaf F on X, the sequence

$$\bigoplus_{i,j} R^{2d}(fu_{ij})_! u_{ij}^* F \rightrightarrows \bigoplus_i R^{2d}(fu_i)_! u_i^* F \to R^{2d} f_! F \to 0$$

is exact.

This is the right exactness of $R^{2d} f_!$, together with the etale descent exact sequence for sheaves established in Expose XVII.

Lemma 2.3. Let $f : X \to S$ be flat and of finite presentation. The locus of X where f is Cohen-Macaulay is relatively of finite presentation and dense in every fibre.

The assertion is local on S, and one reduces to the noetherian case. Fibrewise, over a field, the local rings at maximal points are artinian, hence Cohen-Macaulay. Since the Cohen-Macaulay locus is open, the result follows.

Lemma 2.4. Let $f : X \to S$ be flat, of finite presentation, and Cohen-Macaulay, with S quasi-separated. Every point of X has a Zariski neighbourhood U admitting a quasi-finite flat morphism of finite presentation

$$U \to \mathbb{A}^d_S.$$

This is the local projection result of EGA IV 15.4.3.

Lemma 2.5. Let A be a Cohen-Macaulay local ring of dimension d, with maximal ideal $\mathfrak{m}$. If

$$\underline{u} = (u_1, \ldots, u_d), \qquad \underline{v} = (v_1, \ldots, v_d)$$

are two systems of parameters, then they may be joined by a finite sequence of systems of parameters in which each consecutive pair differs in only one parameter.

The proof is by induction on d. For $d < 2$ there is nothing to prove. For $d \geq 2$, choose $x \in \mathfrak{m}$ avoiding the finite union of associated primes of A/u_1 and A/v_1. Then x is regular modulo both u_1 and v_1, and one applies the induction hypothesis successively in A/u_1, A/x, and A/v_1.

Lemma 2.6. Let X be a Cohen-Macaulay scheme of finite type over an algebraically closed field k, and let

$$u, v : X \to \mathbb{A}^d_k$$

be two quasi-finite flat morphisms. Every closed point $x \in X$ has a Zariski neighbourhood U on which $u|U$ and $v|U$ can be joined by a finite sequence of quasi-finite flat morphisms to $\mathbb{A}^d_k$, each step changing only one coordinate.

After translating the targets, reduce to the case where both morphisms send x to the origin. The germs of quasi-finite flat maps to $\mathbb{A}^d_k$ sending x to 0 are the same as systems of parameters of $\mathcal{O}_{X,x}$. Lemma 2.5 applies. The general case is obtained by first joining u to $u - u(x)$ and v to $v - v(x)$ one coordinate at a time.

If

$$X \xrightarrow{g} Y \xrightarrow{f} S$$

are composable compactifiable morphisms, with relative dimensions $\leq e$ and $\leq d$, respectively, the composition spectral sequence gives by the maximum-cycle argument a canonical isomorphism

$$R^{2d} f_! R^{2e} g_! F \xrightarrow{\sim} R^{2(d+e)} (fg)_! F. \tag{2.7.2}$$

Let $a^d : \mathbb{A}^d_S \to S$ be affine d-space over a coherent base. If F is torsion and prime to the residual characteristics, define recursively an isomorphism

$$\mathrm{Tr}_{a^d} : R^{2d} a^d_! F(d) \xrightarrow{\sim} F. \tag{2.8.1}$$

For $d = 0$ it is the identity; for $d = 1$ it is the trace for curves from Section 1.1; for $d+1$ it is obtained by writing $\mathbb{A}^{d+1}_S = \mathbb{A}^1_{\mathbb{A}^d_S}$ and composing the two traces. This trace is invariant under permutation of the coordinates, because the symmetric group is contained in the connected affine group acting trivially on both source and target after base change.

Theorem 2.9. Let (f, d, F) consist of a compactifiable morphism $f : X \to Y$, an integer d, and a torsion sheaf F on Y prime to the residual characteristics. Assume that f satisfies condition $(*)_d$: there is an open $U \subset X$ on which f is flat of finite presentation with fibres of dimension $\leq d$, and such that $X - U$ has fibres of dimension $< d$. Then there is a unique morphism

$$\mathrm{Tr}_f : R^{2d} f_! f^* F(d) \to F$$

with the following properties.

It is functorial in F; it is compatible with every base change $Y' \to Y$ with Y' quasi-compact and quasi-separated; it is compatible with composition via the isomorphism (2.7.2); and it is normalized as follows: for finite locally free f of rank n and $d = 0$, the composite $F \to f_* f^* F \to F$ is multiplication by n; for $f : \mathbb{A}^1_Y \to Y$ and $d = 1$, the trace is the isomorphism (2.8.1).

For $d = 0$ this is the trace of Expose XVII for quasi-finite flat morphisms. For affine space it is necessarily the trace (2.8.1). Suppose first that f admits a quasi-finite flat map $u : X \to \mathbb{A}^d_Y$. Define

$$t(f, u) = \mathrm{Tr}_{a^d} \circ R^{2d} a^d_!(\mathrm{Tr}_u).$$

This construction is functorial and compatible with base change. To see that it is independent of u, reduce by base change to an algebraically closed field and by Lemma 2.2 to a Zariski-local assertion on X. Lemma 2.6 then reduces to the case where two maps differ in only one coordinate. In that case one factors through $\mathbb{A}^{d-1}$ and uses the invariance of (2.8.1), together with the one-dimensional trace of Section 1.1.

For f flat, Cohen-Macaulay, of finite presentation and pure relative dimension d, choose a quasi-compact open cover on which such quasi-finite flat maps to $\mathbb{A}^d_Y$ exist. Lemma 2.2 glues the local traces to a unique trace Tr_f. For a general f satisfying $(*)_d$, shrink the good open U so that $f|U$ is Cohen-Macaulay and pure of dimension d, and use Lemma 2.1 to identify the top direct image for f with that for $f|U$. This gives existence and uniqueness, together with functoriality, base change, and the normalizations.

It remains only to verify full compatibility with composition. A formal associativity lemma says that, for three composable morphisms $X \to Y \to Z \to T$, if the composition law holds for the first two adjacent pairs, then the two possible triple compositions are equivalent. Thus one reduces to diagrams where both morphisms are described by quasi-finite flat maps into affine spaces. The needed commutativity then follows from the cartesian square

$$\begin{tikzcd} \mathbb{A}^e_S \arrow[r] \arrow[d] & \mathbb{A}^e_T \arrow[d] \\ S \arrow[r] & T \end{tikzcd}$$

with $S \to T$ quasi-finite flat, since after reducing to an algebraically closed field and then to an artinian local algebra, the finite trace is multiplication by the degree while the affine-space trace is stable under base change. This completes the proof of Theorem 2.9.

Proposition 2.10. The trace morphisms of Expose XVII for finite flat morphisms, the trace for smooth curves of Section 1.1, and the affine-space trace (2.8.1) are all special cases of Theorem 2.9.

The trace in Theorem 2.9 is an isomorphism precisely when the geometric fibres of f have exactly one irreducible component of dimension d, with multiplicity prime to n. The notes defer the direct proof, since it follows immediately from the global duality theorem.

Let $f : X \to S$ and $g : Y \to S$ satisfy $(*)_d$ and $(*)_e$. The Kunneth isomorphism gives

$$R^{2d} f_!(\mathbb{Z}/n) \otimes R^{2e} g_!(\mathbb{Z}/n) \xrightarrow{\sim} R^{2(d+e)}(f \times g)_!(\mathbb{Z}/n). \tag{2.11.1}$$

Proposition 2.12. After the appropriate Tate twists, the square

$$\begin{tikzcd} R^{2d}f_!(\mathbb{Z}/n)(d) \otimes R^{2e}g_!(\mathbb{Z}/n)(e) \arrow[r, "\sim"] \arrow[d, "\mathrm{Tr}_f \otimes \mathrm{Tr}_g"'] & R^{2(d+e)}(f \times g)_!(\mathbb{Z}/n)(d+e) \arrow[d, "\mathrm{Tr}_{f\times g}"] \\ \mathbb{Z}/n \arrow[r, "="] & \mathbb{Z}/n \end{tikzcd}$$

is commutative.

This is just the compatibility of the trace with composition, read through the asymmetric construction of the Kunneth morphism used in Expose XVII.

Let $\mathcal{A}$ be a sheaf of rings on S, and suppose $n\mathcal{A} = 0$ with n invertible on S. For $K \in D(S, \mathcal{A})$ there is the projection-formula isomorphism

$$K \otimes^{\mathbf{L}}_{\mathbb{Z}/n} Rf_!(\mathbb{Z}/n)(d) \xrightarrow{\sim} Rf_!(f^*K(d)). \tag{2.13.1}$$

The top trace may therefore be regarded as a morphism

$$Rf_!(f^*K(d)[2d]) \longrightarrow K. \tag{2.13.2}$$

Theorem 2.14. Let $f : X \to S$ be smooth, compactifiable, and pure of relative dimension d, and let n be invertible on S. For every geometric point x of X, with image s in S, there exist an etale neighbourhood V of s and an etale neighbourhood U of x in X_V,

$$\begin{tikzcd} U \arrow[r, "j"] \arrow[dr, "f'_V"'] & X_V \arrow[r] \arrow[d, "f_V"] & X \arrow[d, "f"] \\ & V \arrow[r] & S, \end{tikzcd} \tag{2.14.1}$$

such that the map

$$R^i f'_{V!}(\mathbb{Z}/n) \to R^i f_{V!}(\mathbb{Z}/n)$$

is zero for $i < 2d$, and

$$\mathrm{Tr}_{f'_V} : R^{2d} f'_{V!}(\mathbb{Z}/n)(d) \to \mathbb{Z}/n$$

is an isomorphism.

The proof uses a homological lemma. If $K_0 \to K_1 \to \cdots \to K_{2k}$ is a chain in a bounded derived category, each K_i has cohomology only in degrees $[0, k]$, and each transition induces zero on cohomology in degrees $< k$, then the total composite $K_0 \to K_{2k}$ factors through the top cohomology object $H^k(K_0)[-k]$. Iterating Theorem 2.14 in relative dimension d therefore gives a stronger factorization: after a suitable etale localization, the canonical map

$$Rf'_!(\mathbb{Z}/n)(d) \to Rf_!(\mathbb{Z}/n)(d)$$

factors through $\mathbb{Z}/n[-2d]$ via the trace.

The theorem is proved by induction on d. For $d = 0$ it is tautological, and for $d = 1$ it is exactly the fundamental lemma 1.6.9. For $d \geq 2$, localize so that f factors as

$$X \xrightarrow{h} Y \xrightarrow{g} S$$

with g and h smooth compactifiable of strictly smaller positive relative dimensions d' and d''. Applying the inductive factorization to g and to the base changes of h, one obtains a diagram

$$\begin{tikzcd}
U \arrow[r, "u"] \arrow[dr] & f^{-1}W \arrow[r] \arrow[d] & X \arrow[d, "h"] \\
V \arrow[r, "v"] \arrow[dr] & g^{-1}W \arrow[r] \arrow[d] & Y \arrow[d, "g"] \\
 & W \arrow[r] & S
\end{tikzcd}$$

for which the lower-dimensional trace maps are isomorphisms and the corresponding canonical maps factor through the trace objects. The composition diagram for $Rf_!$, together with the trace compatibility of Theorem 2.9, shows that the composite for f factors through $\mathbb{Z}/n[-2d]$. Hence the maps in degrees $< 2d$ are zero and the top trace is an isomorphism. This completes the proof.